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MSc DATA SCIENCE

Investigating GPT Models for Nutritional Accuracy: A Systematic Review and Performance Enhancement

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Abstract

In the rapidly evolving domain of personalised nutrition, the integration of Artificial Intelligence (AI), especially Large Language Models (LLMs) like the Generative Pre-trained Transformer (GPT) series, offers unprecedented opportunities. This research investigates the capability of GPT models in generating nutritionally accurate dietary recommendations and explores innovative strategies to enhance their reliability and accuracy through the Retrieval-Augmented Generation technique. Initial findings demonstrate a varying degree of nutritional accuracy across different food categories, with certain macronutrients like proteins and carbohydrates presenting challenges in estimation accuracy. To address these limitations, the study introduces a novel approach integrating the NutritionIX API with GPT models to refine the accuracy of generated dietary plans. This integration significantly enhances the precision of nutritional content in AI-generated dietary recommendations up to 90%, aligning them more closely with verified nutritional data. The research not only underscores the potential of LLMs in transforming dietary planning but also highlights the critical role of external databases in enriching AI capabilities, thereby paving the way for more sophisticated applications in personalised nutrition and dietetics. The findings from this study aim to contribute to the broader discourse on the ethical and practical application of AI in healthcare, emphasising the need for rigorous validation and careful integration of AI tools to ensure they enhance public health outcomes without compromising on accuracy or ethical standards.

Keywords: Artificial Intelligence, ChatGPT, Diet, Large Language Models, Nutrition Science, Healthcare, Retrieval-Augmented Generations, National Health Service

1 Introduction

Background

In navigating the complex landscape of public health and individual well-being, dietary guidelines emerge as a critical component, offering evidence-based recommendations to mitigate a host of health concerns, including chronic disease and obesity (King, 2007). The challenge, however, lies in the personalisation of these guidelines to cater to the

diverse needs of the global population, amidst obstacles such as economic disparities, geographic isolation, and social barriers. This gap in accessibility prompts many to turn to the internet, where the reliability of nutritional information often remains questionable, exacerbating the issue of misinformation.

The evolution of Artificial Intelligence, particularly through advancements in Large Language Models like GPT models, introduces a promising avenue for addressing these challenges. LLMs, with their ability to process and generate nuanced information, offer a potential solution to democratise access to personalised dietary advice, available universally at any time (Turchi et al., 2024). This innovation signifies a paradigm shift towards making nutritional guidance more accessible and personalised, leveraging technology to transcend traditional barriers.

However, the application of AI in generating personalised diet plans, especially those that must consider specific dietary restrictions and ensure nutritional accuracy, requires a meticulous approach. Despite the advantages of LLMs in processing vast amounts of information and providing tailored advice, their deployment in healthcare, particularly in nutrition, raises significant concerns regarding accuracy, reliability, and ethics (Meduri et al., n.d.). The complexities of individual health profiles, the importance of precise allergen information, and the critical nature of accurate nutritional advice underscore the need for enhanced strategies to augment the precision and dependability of LLM-generated dietary recommendations.

Recognising these complexities, this dissertation seeks to directly address the challenges inherent in utilising AI for personalised nutrition advice. It aims to develop and validate a strategy that enhance the accuracy and reliability of diet plans generated by LLMs, with a particular focus on ensuring the correctness of nutritional content. By navigating the intersection of AI technology and nutrition science, this research endeavours to advance the field of personalised nutrition. The goal is to leverage the capabilities of LLMs to improve public health outcomes and individual dietary practices through the provision of accessible, accurate, and ethically responsible dietary advice.

Problem Statement

This study addresses the significant challenge of assessing and enhancing the accuracy of nutritional information provided by AI-driven responses, specifically those generated

by GPT models. The rapid adoption of AI in dietary advice necessitates a critical evaluation of the reliability and scientific accuracy of these technologies. Misinformation or inaccuracies in such advice could pose health risks, especially when used to make dietary decisions. This research aims to rigorously assess the current capabilities of GPT models in delivering nutritionally accurate information and to explore advanced strategies, such as Retrieval-Augmented Generation (RAG), to enhance this accuracy.

Research Questions

The study is structured around pivotal research questions to explore the current limitations and potential improvements in AI-driven nutrition advice:

1. What is the current state of nutritional accuracy in responses generated by GPT models?
2. How can RAG be applied to enhance the nutritional accuracy of these models?
3. What are the measurable impacts of integrating RAG into GPT models on the quality and reliability of the nutritional information provided?

Research Objectives

This research sets three clear objectives to systematically tackle the identified challenges:

1. To empirically assess the current state of nutritional accuracy in responses provided by GPT models, identifying key areas of strengths and weaknesses.
2. To develop and implement a RAG strategy aimed at enhancing the accuracy of nutritional information.
3. To evaluate the effectiveness of the RAG-enhanced GPT model through rigorous testing and validation, ensuring that the generated dietary advice aligns with professional dietary standards and real-world applicability.

Significance of the Study

This study seeks to make a substantial contribution to the fields of AI and nutrition by systematically evaluating and enhancing the accuracy of nutrition information provided

by AI systems. By identifying the current limitations of GPT models and implementing advanced techniques like RAG, this research aims to significantly improve the reliability of dietary recommendations, thereby potentially impacting public health positively. Furthermore, by advancing the capabilities of AI in delivering personalised and accurate nutrition advice, this study will provide valuable frameworks and insights that could be broadly applicable in the integration of AI into healthcare.

Ethical Considerations

This research addresses the ethical implications associated with the use of AI in healthcare, particularly in the provision of nutritional advice. It focuses on principles such as ensuring accuracy, safeguarding privacy, maintaining data security, and obtaining informed consent. The introduction of RAG in GPT models will be evaluated not only for its effectiveness but also for its ethical soundness, ensuring that enhancements in AI technology are responsibly implemented. Imaginary user profiles will be used to test hypotheses in a controlled environment, thus maintaining ethical standards and focusing on harm minimization in real-world applications. This methodological choice allows for a comprehensive evaluation within a safe and ethical framework.

2 Literature Review

2.1 Foundations of Nutrition and Health

The journey of nutritional guidelines, rooted in ancient philosophies and religious teachings, has evolved into today's evidence-based recommendations, reflecting a significant deepening of our understanding of diet's role in health, extending beyond mere disease prevention (Schneeman, 2003). This narrative, woven through centuries of scientific inquiry, transitions from early theories like the germ theory (Tomes, 1990) and vitamin deficiency theories (Piro et al., 2010), to contemporary guidance that includes quantitative recommendations for nutrient intake (Board, 1999) and qualitative advice on food choices (Wilson et al., 1979), aimed at enhancing life quality and well-being. Such evolution underscores the responsiveness of nutritional science to emerging research and public health needs, demonstrating a commitment to optimising health through informed dietary practices.

2.1.1 Historical Perspectives on Dietary Guidelines

Transitioning from historical dietary practices, we see a dramatic shift in focus. Initially, the aim was to prevent nutrient deficiencies, especially during challenging times like the Great Depression and World War II (Mozaffarian et al., 2010). Dietary recommendations such as the Recommended Dietary Allowances (RDAs) were formulated to address these deficiencies (Harper, 1987). However, as chronic diseases became more prevalent, dietary guidelines evolved. The 1980 Dietary Guidelines for Americans represent a pivotal moment, introducing a nutrient-focused approach to combat chronic diseases (Nicklas and O'Neil, 2015). This evolution, aligning with global evidence on chronic disease prevention (Organization, 2003, Mozaffarian et al., 2010), marks a significant transition in how we understand and apply dietary guidelines.

2.1.2 Nutritional Science: Principles and Current Understandings

Building on the historical underpinnings of nutritional science, this section delves into the nuanced roles and optimal balances of both macronutrients and micronutrients, emphasising their critical importance in fostering balanced diets and healthful dietary patterns

(Kant, 2004). These considerations form the cornerstone of contemporary nutritional science, reflecting an evolved understanding that moves beyond basic sustenance to optimal health and disease prevention. Carbohydrates, serving as the primary energy source, are recommended to comprise approximately 45-65% of total daily caloric intake. This recommendation underscores the importance of complex carbohydrates, which provide sustained energy, and dietary fibre, essential for digestive health and chronic disease prevention (Burke, 2000, Trumbo et al., 2002). Proteins are fundamental in supporting growth and repair. The dietary guidelines suggest that proteins should constitute about 10-35% of an individual's daily caloric intake. This range accommodates variations in health status, age, and levels of physical activity, highlighting protein's versatile role in the diet (Singh et al., 2017, Trumbo et al., 2002). Fats are delineated as a critical source of essential fatty acids and are advised to account for 20-35% of daily caloric intake. The emphasis is placed on unsaturated fats to support heart health, reflecting the nuanced understanding of fats' roles in diet and disease (Sardesai, 1992, Trumbo et al., 2002). The significance of dietary fibre is also highlighted, with a daily intake of 25-38 grams recommended for adults. This intake is pivotal for promoting digestive health and mitigating the risk of chronic diseases, serving as a testament to fibre's integral role in a healthful diet (Anderson et al., 2009, Trumbo et al., 2002). In addition to macronutrients, micronutrients — encompassing vitamins and minerals — are recognised for their indispensable roles in metabolic processes and disease prevention (Shergill-Bonner, 2017). The establishment of daily values aims to avert deficiencies and uphold overall health, underscoring the meticulous consideration given to micronutrient recommendations within nutritional science. Beyond the individual roles of nutrients, contemporary nutritional science espouses the synergistic effects of food combinations, advocating for diets replete with natural, unprocessed foods to counter chronic diseases (Tapsell et al., 2016). Furthermore, the advent of nutrigenomics heralds a personalised approach to nutrition, offering strategies tailored to individual genetic profiles (Berdanier, 2000). This innovative perspective not only enriches our understanding of diet's impact on health but also signifies a paradigm shift towards precision nutrition, promising avenues for customised dietary recommendations that can optimise individual health outcomes. However, the latter is out of the scope of this study.

2.1.3 The Role of Allergens in Diet Planning

The consideration of allergens in diet planning has become an integral aspect of modern nutritional science, acknowledging the significant impact of food allergies and intolerances on individual health and dietary choices (D’Auria et al., 2019). As the prevalence of food allergies continues to rise globally (Loh and Tang, 2018), understanding and identifying allergens have become crucial in dietary guideline development and personal diet planning. This includes not only recognising common allergens such as nuts, dairy, gluten, and shellfish but also implementing strategies to ensure nutrient adequacy in elimination diets (Groetch, 2013). The challenge lies in balancing the avoidance of allergenic foods with the maintenance of a nutritionally complete diet. This necessitates the inclusion of alternative sources of essential nutrients to prevent deficiencies, underscoring the importance of tailored dietary advice from healthcare professionals (Venter et al., 2012). As nutritional science advances, the focus on allergens reflects a broader shift towards personalised nutrition, aiming to optimise health while accommodating individual dietary restrictions and preferences.

2.2 Healthy and Balanced Diet: Guidelines Based on NHS Recommendations

A healthy and balanced diet is paramount for maintaining good health and preventing chronic diseases. The National Health Service (NHS) in the United Kingdom offers evidence-based guidelines designed to provide individuals with the knowledge necessary to make informed dietary choices NHS (2024a). This section explores the NHS guidelines for a balanced diet, elucidating the scientific rationale behind each recommendation and its potential impact on individuals’ health.

2.2.1 The Concept of a Balanced Diet

A balanced diet fulfils all nutritional requirements of the body without exceeding the necessary intake of calories, thus promoting overall health and preventing nutritional deficiencies and chronic diseases (Lim, 2018). The NHS (2024b) emphasises the importance of including a variety of food items from all major food groups in the right proportions to achieve such a diet. This holistic approach ensures the intake of essential macronutri-

ents—carbohydrates, proteins, and fats—as well as vital vitamins and minerals.

2.2.2 The Eatwell Guide: A Framework for Healthy Eating

The Eatwell Guide is a visual tool developed by the NHS (2024c) to help the public understand the proportions of different food groups required to make up a healthy and balanced diet. As can be seen in the Figure 1, it divides the diet into five key food groups: fruits and vegetables; potatoes, bread, rice, pasta, and other starchy carbohydrates; dairy or dairy alternatives; beans, pulses, fish, eggs, meat, and other proteins; and oils and spreads. The Guide recommends that fruits and vegetables make up over a third of the diet, highlighting the necessity of including a variety of colours and types to benefit from a wide range of nutrients they provide.

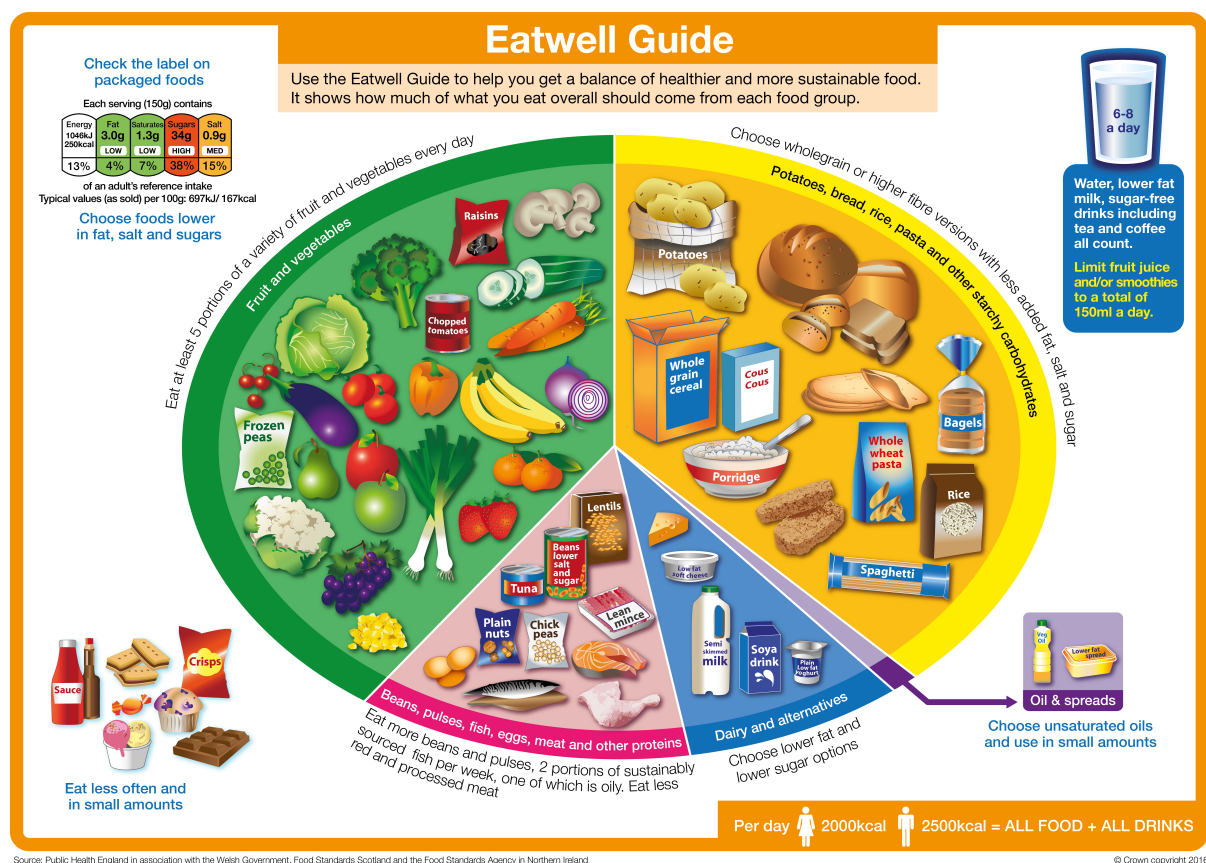


Figure 1: Eatwell guide chart published by UK Government (2024) represents the proportion of different food groups to maintain a healthy and balanced diet.

2.2.2.1 Importance of Fruit and Vegetables: Achieving 5 A Day

A cornerstone of the NHS guidelines is the recommendation to consume at least five portions of a variety of fruit and vegetables every day, known as the "5 A Day" initiative

(NHS, 2024d). This advice is rooted in a wealth of scientific evidence indicating that these foods can significantly reduce the risk of serious health issues such as heart disease, stroke, and some types of cancer (Havas et al., 1994). Fruit and vegetables are rich in vitamins, minerals, and fibre, and are an excellent source of antioxidants, which play a crucial role in protecting the body from damage by free radicals (Kaur and Kapoor, 2001). Strategies to incorporate more fruits and vegetables into the diet include adding fruit to breakfast cereal, snacking on vegetables, and ensuring that a variety of these foods are included in every meal (NHS, 2024d).

2.2.2.2 Starchy Foods and Fibre: The Basis of Meals

Starchy foods, particularly those high in fibre, are recommended by the NHS (2024e) to be the basis of every meal. Foods such as potatoes, bread, rice, pasta, and cereals, especially whole grain versions, provide a significant portion of an individual's energy intake without containing too many calories. High-fibre foods also aid digestion and help prevent constipation, a common dietary issue (Taylor, 1990). They are essential for a healthy gut microbiome, which is increasingly recognised for its role in overall health (Simpson and Campbell, 2015). Practical advice includes choosing wholegrain or higher fibre versions with less added fat, salt, and sugar, and incorporating these foods in varied and enjoyable ways throughout the diet.

2.2.2.3 Protein Sources and Dairy Alternatives

Protein is essential for building and repairing tissues and is a vital component of a healthy diet. The NHS (2024b) guidelines recommend diversifying protein sources to include not only meat and fish but also plant-based options such as beans, pulses, and nuts. This variety ensures the intake of different essential nutrients, including iron, zinc, and B vitamins from meat, and fibre and healthy fats from plant-based sources. Dairy products and their alternatives are highlighted for their calcium content, crucial for bone health, with a recommendation for lower-fat and lower-sugar options to maximise benefits while minimising potential health risks (NHS, 2024f).

2.2.2.4 Fats, Sugars, and Salt: Consumption Recommendations

The NHS (2024b) guidelines advocate for moderation in the consumption of fats, especially saturated fats, sugars, and salt, to reduce the risk of heart diseases, diabetes, and other health conditions. Understanding the types of fats and choosing healthier options like unsaturated fats found in avocados, nuts, and fish can contribute to better cardiovascular health (NHS, 2024g). Recommendations also include reducing the intake of free sugars and salt, with practical advice on reading food labels to make healthier choices and considering natural flavour enhancers like herbs and spices as alternatives to salt (NHS, 2024h).

2.2.3 Hydration: Essential for Health

Hydration is another critical aspect of a healthy diet, with the NHS (2024h) recommending six to eight cups or glasses of fluid a day, including water, lower-fat milks, and unsweetened tea or coffee. Hydration plays a key role in maintaining the balance of body fluids, which are essential for digestion, absorption, circulation, creation of saliva, transportation of nutrients, and maintenance of body temperature (Jéquier and Constant, 2010).

2.2.4 Special Dietary Considerations

In addressing the nutritional considerations of specific dietary patterns, particularly vegetarian and vegan diets, it is essential to ensure these diets are well-planned to meet individual nutritional requirements. Vegetarians, including those who consume dairy and eggs, should aim for a diverse intake of foods to achieve a balanced diet, as recommended by the Eatwell Guide (NHS, 2024i,j). This includes consuming a variety of fruits and vegetables, starchy carbohydrates, particularly whole grains, and multiple protein sources such as beans, pulses, and dairy or fortified alternatives. Key nutrients of focus include calcium and vitamin B12, found in dairy and fortified foods, which are crucial for bone health and metabolic functions, respectively.

Vegans, excluding all animal products, face unique nutritional challenges and must be diligent in consuming fortified foods or supplements to obtain nutrients like vitamin D, vitamin B12, iodine, selenium, calcium, and iron. Adequate protein intake can be

achieved through a variety of plant-based sources, ensuring a mix of essential amino acids. Omega-3 fatty acids, vital for heart health, can be sourced from flaxseed oil, walnuts, and chia seeds among vegans. Additionally, attention should be given to iron, which is less readily absorbed from plant sources, necessitating a diet rich in pulses, whole grains, and green vegetables (NHS, 2024i,j).

Special dietary considerations are paramount during pregnancy and breastfeeding for vegetarians and vegans to support the healthy development of the child. This includes a focus on essential nutrients such as iron, vitamin B12, and omega-3 fatty acids. Children raised on these diets also require careful planning to ensure they receive the necessary energy and nutrients for growth (NHS, 2024k). In summary, with careful planning and consideration of dietary diversity, vegetarian and vegan diets can meet all essential nutritional needs, supporting health and well-being across different life stages.

In conclusion, Section 2.2 has distilled the NHS’s dietary guidelines into a coherent framework, underscoring the scientific basis and practical steps towards a balanced diet. We’ve elaborated on these guidelines to draft a comprehensive prompt template based on health guidelines, which are included in appendix A. These will serve as a foundation in our methodology for generating personalised diet plans, demonstrating the versatility of the guidelines in catering to diverse nutritional needs.

2.3 Digital Health and Personalised Nutrition

The convergence of data science and healthcare technologies has heralded a new era in medical services, where digital health and personalised nutrition stand at the forefront of transforming patient care and health management. This section delves into the crucial role of technological innovations in healthcare, the evolution of personalised nutrition through digital technologies, and the instrumental role of AI in enhancing diet recommendation systems.

2.3.1 Technological Innovations in Healthcare

The digital transformation of the healthcare sector, driven by the advent of technological innovations, has significantly enhanced patient care and operational efficiency (Iyanna et al., 2022). This era is marked by the introduction of varied health information techno-

logies such as telemedicine (Tavares et al., 2023), blockchain applications (Haleem et al., 2021), deep-learning for medical imaging (Maaliw et al., 2022), and AI-powered chatbots (Solanki et al., 2023), revolutionising healthcare through improved clinical decisions and evidence-based practices (Sheikh et al., 2021). The integration of these technologies has been instrumental in reshaping healthcare delivery, promising improved patient outcomes through innovative solutions (Garcia, 2023).

2.3.2 Evolution of Personalised Nutrition through Digital Technologies

The integration of digital technologies has significantly transformed the domain of personalised nutrition, tailoring dietary recommendations to individual genetic, lifestyle, and health profiles (Abrahams and Matusheski, 2020). Digital nutrition applications now offer personalised dietary advice, utilising comprehensive nutritional databases and AI for evidence-based guidance (Abhari et al., 2019). Such applications, including knowledge-based recommender systems like the Virtual Dietitian, leverage extensive dietary knowledge to facilitate informed decision-making, underscoring the potential of digital tools in personalising nutrition (Garcia et al., 2021a).

2.3.3 The Role of AI in Enhancing Diet Recommendation Systems

The emergence of AI-driven solutions marks a significant milestone in the realm of digital healthcare, especially in nutrition (Roychowdhury et al., 2024). Platforms like the Virtual Dietitian represent the evolution of diet recommendation systems into intelligent applications, employing algorithms such as forward chaining to generate personalised meal plans by analysing user profiles, dietary preferences, and restrictions (Garcia et al., 2021). This intelligent reasoning capability, based on extensive databases and inference rules, allows for the customisation of diet plans that optimise nutritional intake while respecting individual dietary needs (Garcia, 2023). The transition towards AI-enhanced systems underscores a broader trend in healthcare, the shift from generic to highly personalised health and nutrition solutions, facilitated by advancements in AI and machine learning technologies.

Building upon the transformative impact of AI in crafting personalised diet recommendation systems, it becomes essential to explore the next frontier in digital health

innovation: the emergence and integration of LLMs, particularly GPT models. These advanced AI constructs offer unprecedented capabilities in understanding, generating, and interacting with human language, heralding a new era of potential in healthcare and nutrition science. The evolution and application of GPT models in these domains not only exemplify the cutting edge of AI technology but also present new opportunities and challenges in delivering personalised healthcare solutions.

2.4 Introduction to Large Language Models and GPT

2.4.1 Evolution of Natural Language Processing: From Basic NLP to GPT Era

The evolution of large language models represents a significant trajectory in the field of AI, particularly in natural language processing (NLP). This journey marks the transition from simple, rule-based systems to the advanced, neural network-driven models of today. The inception of NLP can be traced back to the 1950s and 1960s with the development of basic computational linguistics techniques (Wilks, 2005). Early efforts were focused on rule-based systems where the machine translated text based on a set of manually coded linguistic rules (Shah, 2020). These models were limited by their lack of scalability and inability to understand the nuances of human language.

In the 1980s and 1990s, statistical models began to gain prominence (Wilks, 2005). These models, including hidden Markov models (HMMs) and later, decision trees, utilised statistical inference to predict word sequences and meanings based on large corpora of text (Wu et al., 1992). The statistical approach marked a significant shift towards data-driven methods, laying the groundwork for more complex models.

The advent of machine learning algorithms in the late 1990s and early 2000s revolutionised NLP. Techniques such as Support Vector Machines (SVMs) and later, deep learning, allowed for the automatic extraction of features from text, significantly improving the performance of NLP systems (Nagarhalli et al., 2021). The introduction of word embeddings, such as Word2Vec in the 2010s, represented a paradigm shift, enabling models to understand context and semantic similarity (Jatnika et al., 2019, Zhang et al., 2016).

The implementation of neural networks, particularly Recurrent Neural Networks (RNNs)

and Long Short-Term Memory (LSTM) networks, provided the ability to process sequences of words, capturing temporal dependencies and context (Vennerød et al., 2021). This was a crucial step in developing models that could generate coherent and contextually relevant text.

The introduction of the Transformer architecture in 2017 by Vaswani et al. (2017) marked the beginning of the LLMs era. The Transformer model, characterised by its self-attention mechanism, allowed for parallel processing of sequences, significantly improving efficiency and scalability.

The introduction of the Transformer architecture in 2017 by Vaswani et al. (2017) marked the beginning of the era of modern LLMs. Characterised by its self-attention mechanism, the Transformer model allowed for parallel processing of sequences, significantly improving efficiency and scalability. This architecture facilitated the development of pioneering models like BERT (Bidirectional Encoder Representations from Transformers) by Google, which improved the understanding of context in text by processing words in relation to all other words in a sentence, rather than one direction at a time (Devlin et al., 2018). BERT and similar models like RoBERTa by Liu et al. (2019) and T5 by Raffel et al. (2020) set new standards for model performance across a variety of NLP tasks, creating a robust foundation for future developments.

Building on these foundational innovations, OpenAI introduced its GPT series, which further exemplified the potential of the Transformer architecture. GPT-1 demonstrated unprecedented language understanding and generation capabilities, but it was GPT-2 that showcased the model’s ability to generate coherent and contextually rich text across various domains without domain-specific training (Marcel et al., 2023, Pandey et al., 2024, OpenAI, 2019).

Following GPT-2, OpenAI introduced GPT-3, a more powerful model with 175 billion parameters, significantly advancing the field of NLP. GPT-3 demonstrated remarkable abilities in generating human-like text, solving complex problems, and even programming based on natural language prompts (Kublik and Saboo, 2022). Notably, the model’s capacity to understand and generate text improved dramatically, enabling the development of advanced applications and services. The release of GPT-3.5 further refined these capabilities, enhancing the model’s understanding and interaction quality, thereby setting the stage for even more sophisticated AI-driven conversational agents, such as ChatGPT.

The development of LLMs saw significant advancements with the introduction of GPT-3 and its subsequent versions, which set new benchmarks for performance in various NLP tasks. However, it was the arrival of GPT-4 that marked a pivotal moment in AI research, initiating a shift towards the creation of more generalisable and efficient AI systems, an effort aimed at achieving Artificial General Intelligence (AGI) (Yekta, 2024). This era of GPT models underscored the critical importance of scale—in both the architecture of models and the amount of data used for training—to realise sophisticated language understanding and generation capabilities (Bhattacharjee and Liu, 2024, Marcel et al., 2023).

The transformative journey from basic NLP techniques to the sophisticated capabilities of GPT models underscores not only the vast strides made in AI but also opens up a vista of potential applications within healthcare. As we delve into the realm of GPT in healthcare, it becomes evident that the advanced understanding and generative capabilities of these models offer unique opportunities for enhancing patient care and health management. The ability of GPT models to process and generate nuanced, human-like text makes them particularly suited for tasks such as patient interaction, literature review, and personalised health advice, including dietary recommendations (Haupt and Marks, 2023). This transition marks a critical point where technology meets personal health, bridging the gap between data-driven insights and their practical application in nutrition science. By leveraging the inherent capabilities of GPT models to understand complex dietary needs and preferences, healthcare professionals can tap into a new level of personalised care, aiming to significantly improve health outcomes through tailored nutrition advice. This synergy between advanced NLP capabilities and healthcare applications not only exemplifies the practical utility of GPT models but also highlights the burgeoning field of digital health as a frontier for innovation, promising unprecedented advancements in the way we approach diet and nutrition in healthcare settings (Nag et al., 2024).

2.4.2 Distinguishing Between GPT and ChatGPT Applications in Scientific Literature

The rapid evolution of LLMs like GPT has given rise to a variety of applications, each tailored to meet specific needs and environments. Among these applications, ChatGPT stands out as a specialised iteration of the broader GPT models, designed to facilitate

human-like conversation through a user-friendly chatbot interface. ChatGPT is built upon the architecture and training of the GPT models but is fine-tuned to engage in dialogues and answer queries in a conversational manner. This adaptation enables ChatGPT to mimic human conversational patterns effectively, making it particularly suitable for applications requiring interactive communication. These fine-tuned versions are accessible through the API, allowing developers to integrate ChatGPT's conversational capabilities into their own applications and services (OpenAI, 2024).

In sectors like healthcare specially nutrition science, ChatGPT is often preferred over general GPT models due to its user-friendly chatbot interface. This interface is accessible and practical, especially for health professionals who may lack technical expertise in API integration or coding. Such ease of access has made ChatGPT a popular choice for research and implementation in nutrition science, where professionals often require quick, reliable interactions to obtain dietary information or patient-specific advice.

However, it is important to recognise that the core capabilities of language understanding and generation in ChatGPT are fundamentally the same as those in other GPT models. These capabilities include comprehensive language comprehension, data processing, and text generation. Therefore, findings from studies that employ ChatGPT are generally applicable to other GPT models (OpenAI, 2024). In academic discussions and practical applications within healthcare and nutrition science, ChatGPT and GPT can often be discussed interchangeably in the context of their role in generating medical and nutritional content. In this study, the findings of various academic researches will be reviewed. The majority of them have conducted their research using ChatGPT. However, the model used within ChatGPT in each study is crucial for analysing its efficiency in depth. Therefore, this study will specify the GPT model used within ChatGPT in each study.

2.4.3 GPT in Healthcare: Current Applications and Potential

Recent advancements in GPT models have ushered in a new era of possibilities in healthcare. Notably, iterations such as GPT-3.5 and GPT-4 are transforming medical practice, education, and administration. Muftić et al. (2023) emphasised the profound impact of ChatGPT, highlighting its role in enhancing diagnostic accuracy, improving medical education, boosting patient engagement and satisfaction, and streamlining administrat-

ive tasks. This versatility underscores the transformative potential of GPT models in healthcare.

Building on this foundation, subsequent studies have sought to benchmark GPT models across various healthcare domains. Nógrádi et al. (2023) highlighted that the diagnostic accuracy of an unspecified ChatGPT model nearly matched that of expert neurologists and surpassed that of general medical doctors, showcasing the model’s deep understanding and application of medical knowledge.

Kung et al. (2023) provided further evidence of this capability by testing GPT-3, an earlier version of ChatGPT, which achieved near-passing scores on all three steps of the United States Medical Licensing Examination (USMLE) without specific training on medical content. The study noted the model’s ability to deliver detailed and clinically relevant reasoning, aligning well with the requirements of medical education.

The iterative improvement of GPT models was further documented by Brin et al. (2023), who tested both GPT-3.5 and GPT-4. While GPT-3.5 reached a passing grade of 62.5%, GPT-4 significantly outperformed all previous models with an impressive 90% accuracy rate. Similarly, Giannos (2023) reported that only GPT-4 passed the UK Neurology Specialty Certificate Examination, underscoring the substantial advancements made from GPT-3.5 to GPT-4.

Further benchmarks were set by studies in non-U.S. contexts. Gobira et al. (2023) assessed GPT-4’s accuracy on the Brazilian National Examination for Medical Degree Revalidation, where the model achieved an 87% accuracy rate. However, this study also highlighted GPT-4’s struggles with nullified and complex questions. Another study by Aljindan et al. (2023) focusing on the Saudi Medical Licensing Exam, demonstrated an overall accuracy of 88% for GPT-4, suggesting a high degree of proficiency and a promising future for these models in global healthcare contexts. Additionally In Sun et al. (2023) study , GPT-3.5 and GPT-4 successfully passed the Chinese Registered Dietitian examination.

In addition to diagnostic capabilities, research has also focused on the quality of responses provided by GPT models in various healthcare fields. Yan et al. (2024) examined the quality of GPT-3.5’s responses to dermatology-related queries, noting its effectiveness in supplementing patient education through instant, reliable, and applicable health advice. This model’s utility extends to supporting public health initiatives as highlighted by

Biswas (2023), who recognised GPT-3.5 as a valuable tool for disseminating information on health topics, aiding disease prevention strategies, and promoting health programs. However, Biswas also cautioned against potential limitations such as data biases, a lack of contextual understanding, and the absence of direct interaction with healthcare professionals, stressing the importance of integrating these AI tools with human expertise for effective use.

Further exploring the utility of AI in clinical settings, Liu et al. (2023) reported on a study where five clinicians assessed the quality of AI-generated suggestions for clinical decision support systems. The clinicians found these suggestions to offer unique perspectives and deemed them highly understandable and relevant. Despite the moderate usefulness and low acceptance rates due to perceived biases, inversions, and redundancies, the study underscored the potential of AI-generated suggestions as a complementary resource for optimising CDS alerts, improving alert logic, and supporting the implementation of enhancements.

Johnson et al. (2023) conducted a broader evaluation of GPT-3.5’s medical responses across various fields. Their findings suggested that while the integration of language model applications like ChatGPT shows early promise in medical practice, there are significant considerations for ensuring their safe and optimal use. Despite high accuracy and completeness scores across various specialties, question types, and difficulty levels, the study concluded that further development is essential to enhance the reliability and robustness of these tools before they can be integrated into clinical practice.

Another study by Gravina et al. (2024) explored the application of GPT-3.5 in the field of inflammatory bowel diseases (IBD), noting the model’s potential despite significant limitations in updating details and occasionally providing inaccurate information. The study suggests that further refinement of ChatGPT, possibly by aligning outputs with leading medical evidence from reliable databases, is necessary for its effective use in healthcare. In a related study, Samaan et al. (2023) assessed the performance of GPT-4 in the same medical field. They found that GPT-4 provided comprehensive responses to the majority of IBD-related questions, underscoring the potential of LLMs as supplementary tools for patients seeking nutrition-related information. Nonetheless, with 17% of responses containing incorrect information, the study highlighted the crucial need for continuous refinement and validation of LLMs before they are integrated into clinical

practice.

Additionally, Dhar et al. (2024) evaluated GPT-3.5’s effectiveness in addressing common questions from patients and caregivers following a tonsillectomy. While the model often provided well-articulated responses, it also risked presenting inaccurate information as factual, raising concerns about safety and reliability.

Beyond the performance qualification of these models, Adams et al. (2023) examined GPT-4’s capability to structure free-text in radiography reports without compromising accuracy. Their findings suggest promising applications for developing smart data pipelines that can transform conversational outputs into database-readable formats, potentially enhancing data management in medical systems.

The studies reviewed illustrate both the substantial potential and the inherent challenges of employing GPT models in healthcare. While these models can significantly enhance information accessibility and support patient education, their current limitations in accuracy and reliability necessitate ongoing development and rigorous validation. As these tools evolve, they must be carefully integrated with existing medical knowledge bases and clinical workflows to ensure they contribute positively to patient outcomes.

Moving forward, this study will explore these advanced models’ performance in the field of personalised nutrition, reflecting a crucial area of potential in leveraging AI for individualised healthcare solutions.

2.5 LLMs in Personalised Dietary Advice

2.5.1 Capabilities of GPTs for Nutrition Science

The capabilities of advanced language models like GPT-3 and GPT-4 in nutrition science are just beginning to be explored, highlighting a promising area for future research given the importance of nutrition to overall health. A study by Kirk et al. (2023) notably evaluated GPT-3 against professional dietitians, revealing that ChatGPT outperformed human experts in five out of eight categories, including overall scores, scientific accuracy, actionability, and comprehensibility. This indicates that such models could effectively handle nutrition-related inquiries, demonstrating potential utility in providing nutrition assistance.

Further research into the accuracy of these models, such as the work by Haman et al.

(2024), shows that 97% of energy values generated by GPT-3.5 was within a 40% range of United States Department of Agriculture (USDA) data, with consistently low variation in nutrient content, and Table 1 represents statistical significance of GPT-3.5 responses in details. This was supported by Lo et al. (2023) who confirmed the alignment of GPT-4’s responses with USDA standards. On the other hand, in Niszczoła and Rybicka (2023) study it is highlighted that energy values were inaccurately calculated and misaligned with their respective prompts in some cases. Furthermore, computational errors were found in the energy values of certain food items and meals. Consequently, the acceptance of AI-generated nutritional advice within the scientific community remains under debate, as there highlighting the need for ongoing scrutiny and validation.

Hoang et al. (2023) compared the performance of GPT-3.5 and GPT-4 in generating nutritional information, finding little difference between the two models. However, GPT-4 was noted to occasionally overestimate the protein content in foods, suggesting areas for model refinement. These studies collectively underscore the potential of GPT models in nutrition science but also emphasise the necessity for meticulous development and evaluation to ensure their reliability and accuracy in practical applications.

Table 1: Accuracy of GPT-3.5 nutritional information within specified bounds(Haman et al., 2024)

Nutrient	Bound 10 (%)	Bound 20 (%)	Bound 30 (%)	Bound 40 (%)	Bound 50 (%)
Carbohydrate, by difference	52.31	65.00	76.35	80.96	85.96
Energy	66.39	80.21	88.66	96.29	96.91
Protein	48.18	71.73	84.53	90.76	94.49
Total lipid (fat)	29.87	47.67	58.77	69.69	75.68
Water	62.02	73.95	79.74	82.46	84.21

In a comparative study by Papastratis et al. (2024), the performance of GPT-3.5 and GPT-4 models was measured against state-of-the-art Knowledge-Base (KB) food recommender systems. According to Figure 2, KB systems consistently surpassed the GPT models in the accuracy of nutritional information, particularly for specialised cases such as specific diseases. It was observed that providing individualised calorie intake data enhances the accuracy of GPT models. Furthermore, GPT systems are capable of offering a more diverse range of meal recommendations compared to KB systems, an attribute that proves beneficial to dietitians and individuals seeking varied dietary

options. Nevertheless, the capability of these models to provide balanced meal plans requires further enhancement.

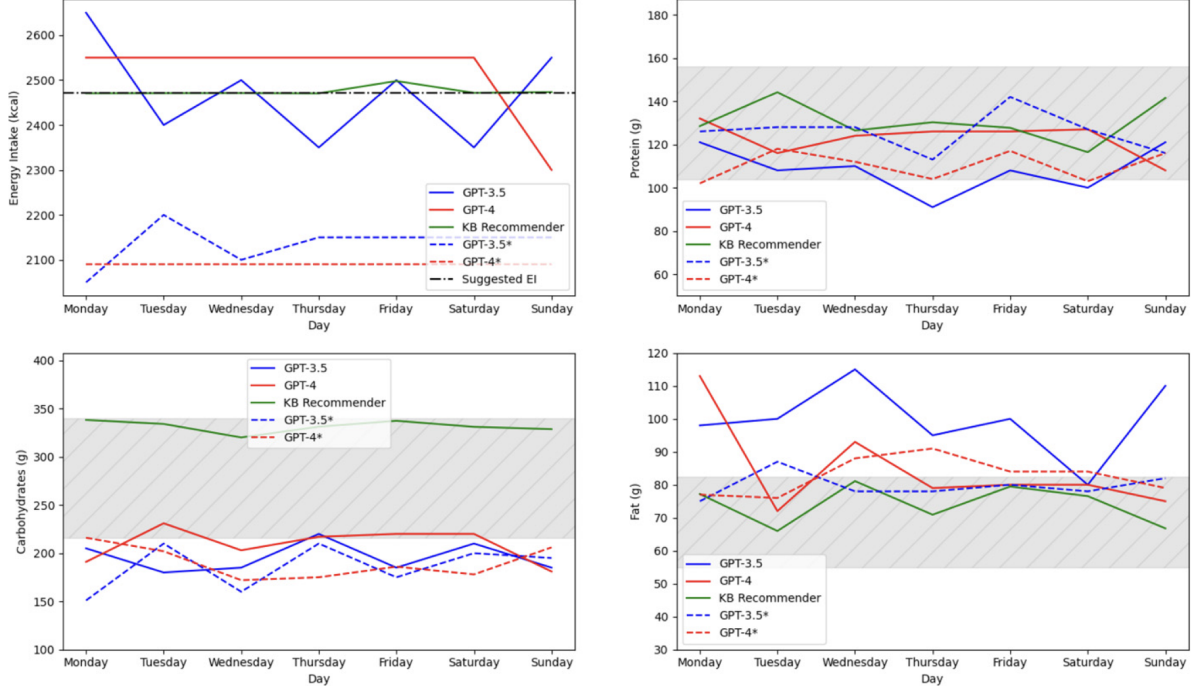


Figure 2: Weekly meal plan statistics for an obese patient using GPT-3.5, GPT-4, and the KB recommender. Grey zone represents the acceptable range of values for the respective nutrient (i.e., carbohydrates, protein, fat). GPT-3.5* and GPT-4* denote the GPT instances in which the personalised target energy intake was not given at the input level. EI, energy intake; KB, knowledge-based (Papastratis et al., 2024).

Further research has been conducted to integrate GPT models into healthcare, specifically for managing diet in patients with type 2 diabetes. Sun et al. (2023) reported that dietary suggestions from GPT-3.5 largely adhered to recognised best practices, although there were exceptions, such as recommendations involving root vegetables and dry beans. In addition, a majority of professional dietitians (96.4%) reviewed GPT-3.5’s answers to common queries positively, indicating a high level of endorsement. In related research, Barlas et al. (2024) noted that GPT-3.5 not only provided consistent responses but also recommended consulting healthcare professionals for personalised advice, particularly in complex cases like obesity management in type 2 diabetes. The model’s responses aligned perfectly with established guidelines in general discussions about obesity, yet exhibited variability in advice concerning nutritional, medical, and surgical weight loss treatments.

These studies collectively demonstrate the proficiency of GPT models in offering nu-

tritional information and dietary advice. However, the challenge remains in crafting tailored diet plans that consider intricate details such as macronutrient ratios, micronutrient levels, suitable food combinations, and allergen information.

2.5.2 GPTs Performance in Generating Diet Plans

Recent studies have extensively evaluated the capabilities of LLMs in generating diet plans, demonstrating their potential for both general and clinical applications. Kim et al. (2024) provided evidence that diet plans generated by GPT-4 are indistinguishable from those crafted by human experts, highlighting the model’s advanced understanding and application of nutritional knowledge. Similarly, Haman et al. (2024) found that GPT-3.5 could produce daily meal plans adhering to specific caloric limits, maintaining meals within 30% of USDA calorie guidelines as detailed in their Figure 3. These findings suggest that GPT-3.5 consistently delivers accurate nutritional information within meal plans.

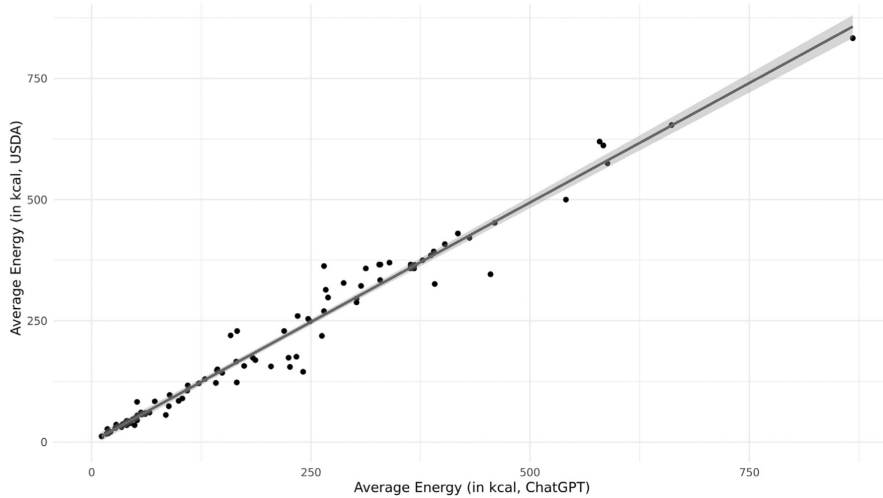


Figure 3: Comparison of energy values from the United States Department of Agriculture and GPT-3.5 (Haman et al., 2024).

Further research by Podszun and Hieronimus (2023) assessed the nutritional completeness of meal plans created by GPT-3.5 across omnivorous, vegetarian, and vegan diets. While the plans typically exceeded protein recommendations, they often lacked adequate amounts of carbohydrates and fats. Although saturated fat and fibre levels were within acceptable limits, discrepancies in micronutrient adequacy were noted, with

details provided in Figure 4. These results underscore the need for human oversight to adjust AI-generated meal plans to fully meet nutritional standards.

Expanding on the theme of personalisation, Papastratis et al. (2024) demonstrated that by integrating additional personal data, GPT models can produce more tailored and precise dietary recommendations, significantly enhancing their utility in personalised dietary guidance. This ability to customise outputs is crucial for addressing individual dietary needs and preferences.

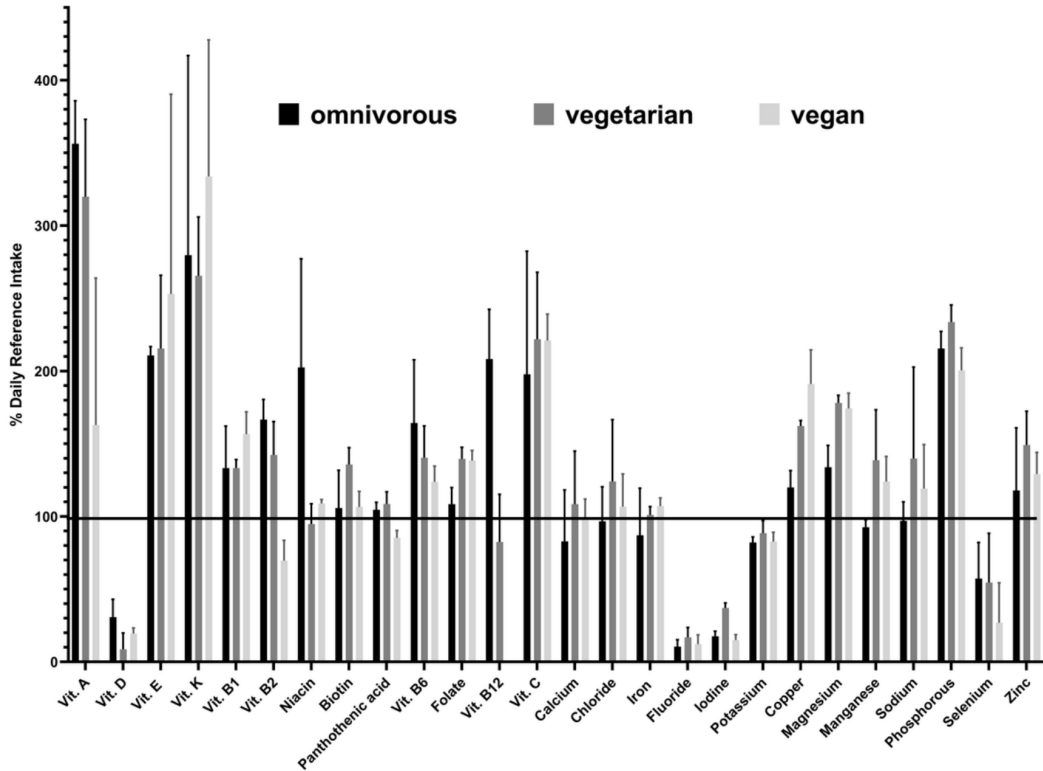


Figure 4: Average micronutrient content of the GPT-3.5 generated meal plans for omnivorous, vegetarian, and vegan diets (Podszun and Hieronimus, 2023).

Dergaa et al. (2024) further investigated the application of GPT-3.5 in creating diet plans tailored to specific health conditions, offering a variety of meal options that comply with evidence-based guidelines. Despite the model’s ability to cater to individual preferences and dietary restrictions, there were notable limitations in its accuracy concerning macronutrient distribution, management of health conditions, drug interactions, and setting realistic weight loss goals.

Lastly, Mishra et al. (2024) conducted a comparative analysis of GPT-3.5’s nutrition recommendations across various medical scenarios. As depicted in Figure 5, their study

revealed that while the model’s response accuracy was consistent, the complexity of medical cases was positively correlated with potential risks. This indicates that while LLMs like GPT-3.5 can handle increasing complexity, they also introduce greater risk of inaccuracies in more complex medical contexts.

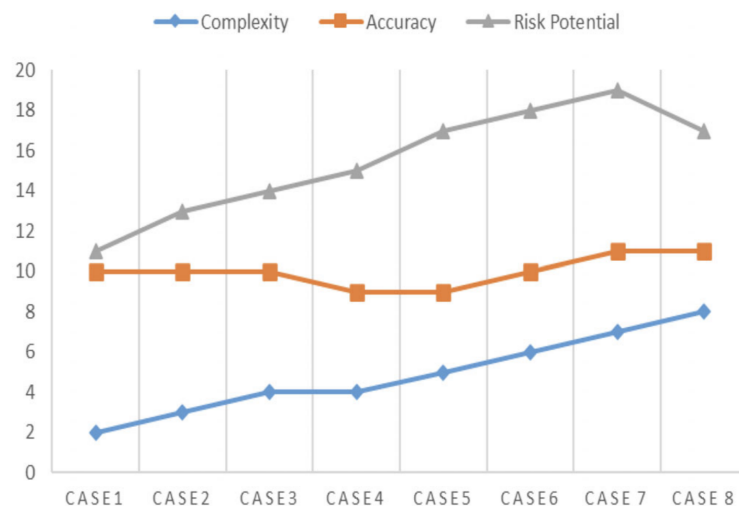


Figure 5: Comparative Analysis of Complexity, Accuracy, and Risk Potential Across Eight Cases (Mishra et al., 2024).

In conclusion, the studies reviewed consistently demonstrate the capability of large language models like GPT-3.5 and GPT-4 in generating nutritionally informed meal plans. While these models can closely mimic human-like precision in general scenarios, their application in clinical and personalised diet planning still requires careful oversight. Although they exhibit strong potential for customising diet plans based on individual data, limitations in nutrition information accuracy, micronutrient balance, specific health condition management, and complex medical case handling necessitate further refinement. Hence, while LLMs represent a promising tool for dietary planning, their integration into clinical practice should be approached with a balanced perspective, emphasising both their strengths and areas for improvement.

2.5.3 Challenges and Limitations

In examining the potential and limitations of AI tools such as ChatGPT in the context of nutrition and dietary planning, a body of recent academic research provides a multifaceted view. Studies indicate that while ChatGPT can augment nutrition knowledge and assist

in dietary education, it is not a substitute for professional dietitians or nutrition experts. For instance, Garcia (2023) suggests that the use of ChatGPT should be supplementary, emphasising the need for careful implementation and oversight to guarantee safety and effectiveness. This call for a cautious approach is echoed in further studies that highlight the limitations of AI in managing complex health conditions.

Ponzo et al. (2024) notes that although GPT-3.5 can provide fairly accurate nutritional advice for various non-communicable diseases, it struggles with complex cases that involve multiple health conditions, occasionally delivering contradictory or inappropriate advice. This underscores that AI, at its current stage, cannot replace the nuanced, expert advice of healthcare professionals, particularly in scenarios that require tailored dietary strategies.

The risks associated with AI-generated dietary plans are further discussed in studies focusing on the accuracy of such tools. Niszczota and Rybicka (2023) found that although most AI-generated diets correctly excluded specified allergens, errors were still prevalent, such as the inclusion of allergens in a small number of cases and inaccuracies in food portions or calorie counts. This highlights significant safety concerns, particularly for users with specific dietary needs, and suggests a pressing need for improvements in AI technologies and their oversight.

Mishra et al. (2024) and Sun et al. (2023) both stress the importance of integrating AI as a decision support system rather than a standalone decision-maker in clinical settings. They advocate for policies that ensure all AI-generated information is verified by human professionals within a short timeframe, enhancing healthcare efficiency while safeguarding patient safety.

Finally, Dergaa et al. (2024) and Barlas et al. (2024) highlight a critical issue: patients and individuals should not rely solely on AI for dietary planning due to inherent risks and the often insufficient responses that require further prompts to align with established guidelines.

In summary, while AI like ChatGPT has shown promise in supporting dietary education, its integration into clinical practice and personal health management must be handled with extreme caution. These studies collectively emphasise the need for continued research, careful policy-making, and rigorous oversight to optimise the benefits of AI while mitigating its risks.

2.6 Discussion and Conclusion of Literature Review

The review of current literature reveals a nuanced landscape regarding the performance of LLMs, such as GPT, in nutrition science which is summarised in Figure 6. These models exhibit notable strengths that make them potent tools for enhancing dietary planning. Among their advantages, they generate diverse diet plans, which may boost adherence and overall satisfaction with dietary regimens. Moreover, their ability to consistently provide energy values and generally sufficient micronutrient information except for some specific minerals underscores their potential as supportive tools in dietetics.

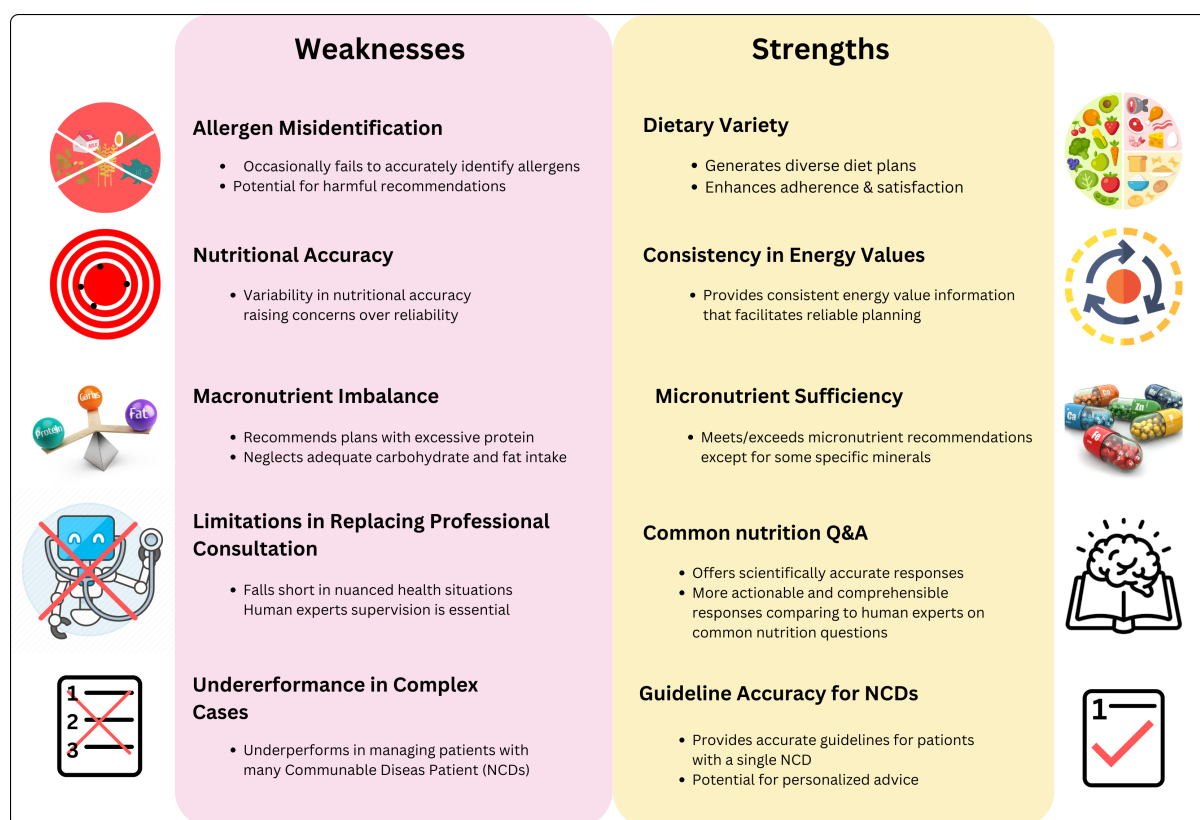


Figure 6: Strengths and Weaknesses of GPT-3.5 and GPT-4 in the field of nutrition science

However, the analysis also highlights critical weaknesses that necessitate cautious integration of these models into nutritional planning. The occasional misidentification of allergens and the variability in nutritional accuracy are significant concerns, as these can lead to harmful recommendations, particularly for individuals with specific health conditions. Furthermore, LLMs often recommend diet plans with an imbalance in macronutrients, typically overemphasising protein while neglecting adequate carbohydrate and fat intake. This skewed macronutrient distribution could potentially lead to unbalanced diet

plans which do not align with established dietary guidelines.

The literature also indicates that while LLMs can handle common nutrition queries effectively and provide guidelines for patients with straightforward conditions, they underperform in complex cases, such as managing patients with multiple non-communicable diseases. This limitation is compounded by their inherent inability to replace professional consultation, particularly in nuanced or critical health scenarios requiring human expert judgement.

The critical review of these models reveals a significant research gap in the reliability and accuracy of nutritional information provided by LLMs. Given that nutritional accuracy is paramount in diet planning and can significantly impact individual health outcomes, addressing these deficiencies is crucial. Therefore, this study will focus on evaluating the current state of nutritional accuracy in GPT model responses and exploring how the infusing the external databases into LLMs can specifically address these limitations. By concentrating on this research gap, the aim is to refine the capabilities of LLMs, making them more reliable and effective tools in personalised nutrition planning. This approach not only promises to advance the field of nutrition but also contributes to the broader discourse on the ethical and practical application of AI in healthcare.

As this study transitions from the literature review to the methodology section, the focus will now shift towards designing an experimental framework to rigorously assess and enhance the nutritional accuracy of GPT models, thereby aiming to fill the identified research gap and potentially revolutionise the application of LLMs in dietetics.

3 Methodology

The methodology of this study comprises two principal experiments designed to enhance and assess the accuracy of nutritional information generated by GPT. The initial stage involves a comprehensive analysis of the accuracy of nutrition values provided by GPT models, utilising a series of predefined metrics to evaluate performance. Subsequently, the second stage introduces a novel method aimed at refining the nutritional accuracy of LLM responses, with its effectiveness tested in real-world scenarios related to diet planning.

To facilitate this research, the NutritionIX database was integrated through its API, serving as the benchmark for accurate nutritional information (NutritionIX, 2024). Previous studies, such as those conducted by Ho et al. (2020), Dorsch et al. (2020), Chikwetu et al. (2023), Steinberg et al. (2020), have utilised the NutritionIX API for similar purposes.

NutritionIX, an NLP-based database, interprets user queries, such as food item names, and retrieves comprehensive nutritional data. This database uniquely addresses the challenge of name variations—for instance, 'chia seed', 'chiaseed', and 'chia-seed' are recognised as identical entities. This feature is particularly valuable, as LLMs often generate diverse nomenclature in diet plan outputs, thus the API's ability to reconcile these variations is critical. The efficacy of this functionality will be rigorously tested in the second experiment.

3.1 First Experiment: Accuracy of Nutritional Information Generated by GPT

To evaluate the accuracy of models in estimating nutritional information, this study undertook a detailed analysis of 150 common food items and ingredients. These items were categorised into five groups based on the NHS Eatwell Guide: Starchy Carbohydrates; Fruits and Vegetables; Dairy and Alternatives; Beans, Fish, Eggs, Meat, and other Proteins; and Oils and Spreads. Each category was assigned 30 food items, forming the foundation for a comparative evaluation across groups.

3.1.1 Dataset Synthesis

Accurate dataset synthesis is crucial for analysing the performance of LLMs in nutritional information estimation. This study generated five distinct datasets, each designed to explore different facets of model behaviour and sensitivity to input variations. These datasets were created to ensure a comprehensive assessment of how model configurations, such as changes in input order and model temperature, impact the accuracy of the nutritional data produced.

Dataset 1: Macronutrient Values The baseline dataset comprises the actual macronutrient values (energy in kcal, protein, carbohydrates, fats, and fibre in grams) for each item. This data was sourced using the NutritionIX API.

Dataset 2: GPT-3.5-Turbo Synthesis Utilising gpt-3.5-turbo-0125 at a temperature setting of 0, this dataset was generated to simulate nutritional values. The model was queried five times per ingredient, with randomised order of ingredients each time to mitigate prompt order bias. The specific prompt used is included in appendix B.

Dataset 3: GPT-4-Turbo Synthesis This dataset mirrors the conditions of Dataset 2 but employs GPT-4-Turbo-2024-04-09, allowing for a comparative analysis between model versions under similar test conditions.

Dataset 4: Non-Shuffled Ingredient Order In contrast to Dataset 2, the order of ingredients in this dataset was kept consistent across all five model queries to determine the impact of ingredient sequence on the generated nutritional information.

Dataset 5: Variation in Temperature Using the same setup as Dataset 2, this dataset was synthesised at a temperature of 0.7—commonly used in chatbot applications—to examine the effect of model hyperparameters on output accuracy.

Recognising the challenges posed by long contexts in LLMs, as noted by Li et al. (2024) regarding their compromised reasoning capabilities, each dataset was generated by dividing the full list of foods into smaller segments. This segmentation ensured that the model only needed to generate information for a manageable number of items at a time, thus preserving the accuracy of the responses. The models’ outputs, structured as JSON strings, are exemplified in appendix C. All generated outputs were subsequently merged into a uniform dataframe for analysis.

Table 2: Description of Datasets and Models

Dataset Number	Description	Model Used	Temperature	Order Variation	Purpose
Dataset 1	Macronutrient Values	NutritionIX API	N/A	N/A	Provides actual nutritional values for baseline comparison.
Dataset 2	GPT-3.5-Turbo Synthesis	GPT-3.5-Turbo-0125	0	Randomized	Tests model accuracy under controlled temperature with order of ingredients randomized.
Dataset 3	Non-Shuffled GPT-4-Turbo Synthesis	GPT-4-Turbo-2024-04-09	0	Consistent	Allows comparative analysis between model versions under similar conditions.
Dataset 4	Non-Shuffled Ingredient Order	GPT-3.5-Turbo-0125	0	Consistent	Investigates the impact of consistent ingredient order on model output.
Dataset 5	Variation in Temperature	GPT-3.5-Turbo-0125	0.7	Randomized	Examines the effect of temperature variation on model output accuracy.

3.1.2 Implications for Research

This study’s methodical approach enables a systematic investigation into the performance of LLMs in estimating nutritional information. By employing different configurations of LLMs, this research could illuminate critical insights into both the potential limitations and areas for optimisation of these models within the context of nutritional data estimation. Specifically, the categorisation of foods into distinct groups according to the NHS Eatwell Guide allows for a nuanced analysis that assesses whether model performance varies across different food categories. Such findings may indicate the need for tailored model training or adjustment of parameters based on food type, thereby enhancing the accuracy and reliability of nutritional estimations provided by LLMs. This could have significant implications for applications in health, dietetics, and food service industries, where precise nutritional information is crucial for decision-making and planning.

3.1.3 Statistical Analysis

To accurately evaluate the precision of nutritional information provided by each model, this study will utilise several statistical methods. We’ll calculate the Mean Absolute Error

(MAE) and Root Mean Squared Error (RMSE) to quantify the discrepancies between the predicted and actual values of macronutrients within similar food groups.

Given that the concentration of nutrients varies significantly across different types of food—for example, starchy foods generally contain higher carbohydrate levels than fats, and meats may have a greater proportion of proteins relative to carbohydrates—it’s crucial to employ a metric that normalises these variations. The Mean Absolute Percentage Error (MAPE) is typically used for this purpose. However, MAPE can be problematic when dealing with food items that contain zero content of a specific macronutrient (e.g., fibre), leading to division by zero errors. To overcome this issue, this study will use the Symmetric Mean Absolute Percentage Error (SMAPE). SMAPE adjusts errors to a scale of 0-200%, facilitating the comparison of errors across different macronutrient groups and resolving the division by zero issues associated with MAPE.

3.2 Second Experiment: Enhancing Nutrition Accuracy Using Retrieval-Augmented Generation and NutritionIX

To enhance the precision of nutritional data within diet plans, this study introduces a novel methodology employing the RAG technique. This approach markedly improves the accuracy and reliability of LLMs by dynamically incorporating domain-specific data from external databases, thereby proving invaluable for knowledge-intensive tasks such as dietary planning (Gao et al., 2023). Similar method has been utilised previously by Abbasian et al. (2024). This study specifically targets the reduction of nutritional inaccuracies by integrating data from the NutritionIX database into LLM-generated responses, thereby refining the dietary recommendations provided.

3.2.1 Retrieval-Augmented Generation Technique

Retrieval-Augmented Generation is a hybrid technique introduced by Lewis et al. (2020) that enhances LLMs by integrating retrieval mechanisms with generative capabilities. This approach leverages both a pre-trained language model and an external knowledge retriever to dynamically fetch relevant documents or data during the generation process. The RAG model works by querying a dataset of documents, extracting contextually pertinent information, and using this information to inform the generation of responses.

This method enriches the model’s output, ensuring it is not only contextually aware but also factually accurate and up-to-date.

For applications in meal planning, RAG can significantly enhance the factual accuracy of nutritional information. By retrieving the latest data from trusted nutritional databases in real-time, RAG ensures that the generated meal plans are both accurate and tailored to current dietary guidelines and scientific findings. This capability is especially valuable for creating personalised meal plans that adapt to evolving nutritional science and individual dietary needs.

3.2.2 Experiment Design

The methodology unfolds in several structured phases:

1. User Profile Initialisation: To generate a diverse array of food items for this experiment, the study created 20 randomised user profiles, ensuring varied dietary patterns to reflect a broad demographic spectrum. These profiles included both male and female genders and spanned ages from 18 to 70. Activity levels ranged from minimal activity to professional athleticism. Dietary preferences incorporated were vegan, vegetarian, and omnivorous patterns. Health objectives such as fat loss, muscle gain, and maintenance were also considered. The profiles varied in weight (40 to 120 kg) and height (150 to 200 cm), and included specific food allergies—nuts, milk and dairy, gluten, and eggs. After establishing these parameters, the study employed ChatGPT to simulate controlled user profiles, which mirrored a wide range of demographic characteristics. This methodology not only ensured the inclusion of a diverse set of food items in the diet plans but also addressed ethical considerations regarding the use of real human profiles. It is pertinent to mention that the primary focus of this study is not the precise alignment of these diet plans with the user profiles based on health protocols, but rather to ensure that the food items cater to a broad spectrum of dietary needs and preferences.

2. Diet Plan Generation: To create personalised diet plans, the language model (LLM) is queried using user profiles that integrate a range of parameters: age, gender, physical attributes, dietary restrictions, and specific health goals. These profiles are synthesised into prompt templates, which are crafted based on NHS guidelines to ensure scientific validity which can be found in appendix A. The finalised prompt is then used to query the GPT, facilitating the generation of individualised diet plans. This method

ensures that the diet plans are not only personalised but also adhere to scientifically reviewed health standards, encompassing a diverse array of food items. Figure 7 illustrates this process, depicting the integration of user profiles with prompt templates and the querying of the GPT to produce the diet plans.

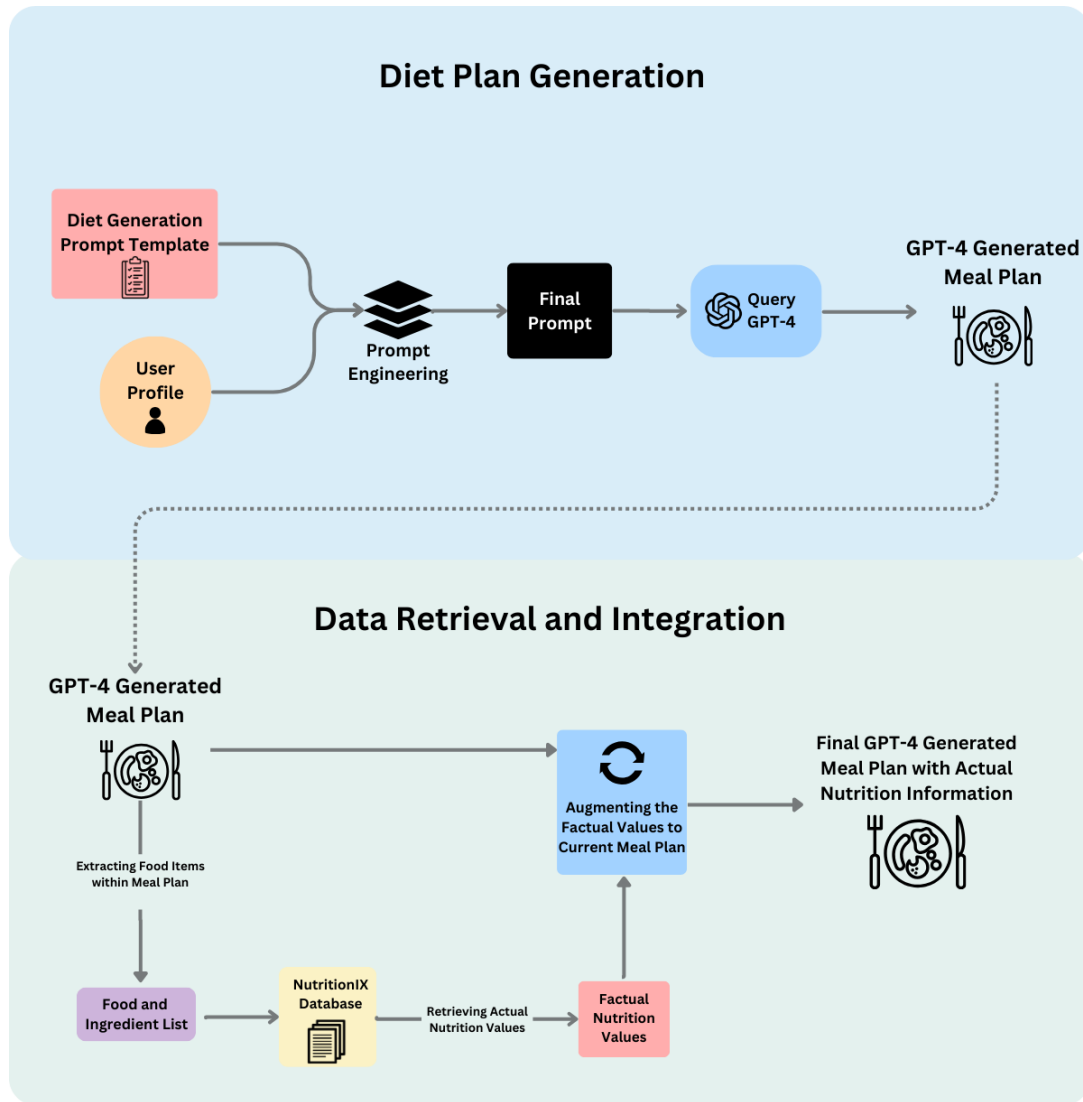


Figure 7: The process of initialising diet plans, retrieving food items, and augmenting nutritional values.

3. Data Integration and Retrieval: Each diet plan generated by the Language Learning Model (LLM) is followed by a JSON formatted version, which systematically structures the meal data to facilitate further processing. An example of the system output can be found in Appendix D. This structured data format simplifies the linkage of dietary information with external databases. Once the food and ingredient list is extracted from the meal plan JSON, the items within this list are passed to the NutritionIX API to

retrieve accurate nutritional details. An example of a food and ingredient list is available in Appendix E.

4. Data Update Using RAG: Post-retrieval, the nutritional values within the meal plan’s JSON are updated to reflect the accurate data obtained from NutritionIX. This revised JSON is then re-input into the LLM using the RAG technique, enabling the model to recalibrate the meal plan based on the enriched nutritional data. This refinement step is crucial for minimising the discrepancies in nutritional information initially produced by the LLM. Figure 7 displays data retrieval and integration visually.

To rigorously evaluate the efficacy of this solution, a comprehensive experimental design has been established. This experiment aims to evaluate the performance and applicability of this methodology in real-case scenarios. Profiles for 20 hypothetical users were generated, each including attributes such as age, gender, weight, height, activity level, dietary restrictions, health goal, and daily calorie intake. These profiles serve as the basis for the generation of personalised meal plans. They were used to craft meal plans that adhere to NHS guidelines for a balanced diet, details of which are reviewed in the literature.

Real-Case Diet Generation: For the production of highly realistic and accurate diet plans, GPT-4-Turbo-2024-04-09 was selected due to its enhanced ability to follow complex instructions. This model ensures higher fidelity in the simulation of real-world diet scenarios. Using the constructed profiles, a template for diet generation that adheres to NHS guidelines for a balanced diet was developed, as reviewed in the literature and detailed in appendix A.

In contrast to the manually selected food items in the first experiment, the meal plans in this experiment are designed to reflect the diverse and realistic choices found in actual diets. An example of a meal plan generated by GPT-4-Turbo, including the macronutrient content for each item, is provided in appendix D.

Dataset Creation and Analysis: From the outputs generated by GPT, a food dataset was compiled, similar to Dataset 2 from the first experiment. The names of these food items were then used to query the NutritionIX API, creating a new dataset that mirrors the approach used for Dataset 1 in the first experiment. This process allows for the assessment of the accuracy and completeness of the nutritional data retrieved by NutritionIX, providing insights into its effectiveness.

Exploratory Data Analysis: To evaluate the accuracy of the nutritional information provided by NutritionIX, exploratory data analysis (EDA) will be conducted. This analysis will determine the feasibility of enhancing nutrition information of food items generated by the GPT model. The results of this EDA will be instrumental in assessing the utility of external knowledge for practical applications in dietary planning and nutrition management.

4 Findings and Analysis

4.1 Accuracy of Nutritional Information Generated by GPT

In the first experiment, this study employs a structured approach to assess the capabilities of GPT-3.5 and GPT-4 models across four different settings, each involving the generation of nutrition information for a set of 150 food and ingredient items. These items are categorised into five groups based on NHS guidelines. Initially, the study compares the performance of the GPT models across these settings to identify the most accurate configuration. The selected setting is then subject to further detailed analysis. This includes a thorough evaluation of the accuracy of macronutrient data generated by the GPT models. Additionally, the study examines the models' performance within each of the five food categories, ensuring that the research not only assesses overall macronutrient estimation accuracy but also identifies specific categories that may require performance enhancements. The findings are intended to guide future research, highlighting areas of weakness and recommending opportunities for improvement.

4.1.1 Comparative Analysis of Model Settings' Effects on LLM Nutritional Accuracy

4.1.1.1 Input Sequencing

In the development of Datasets for the study, a specific querying methodology was employed using a LLM. Each food item was queried five times, with each query comprising a list of various ingredients that were submitted to the LLM. For populating dataset 2, the order of ingredients was randomised for each of the five queries. Conversely, in a comparative series, the order of ingredients was kept consistent across all queries for dataset 4.

Table 3: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Using Shuffled Input Sequencing and Zero Temperature on Dataset 2 Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE
Fiber	1.700	2.024	0.637	2.072
Protein	8.330	8.354	1.263	2.813
Fats	22.094	21.996	0.874	1.759
Energy	281.312	286.373	18.213	46.628
Carbs	13.942	15.213	3.022	8.635

Table 4: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Using Consistent Input Sequencing and Zero Temperature on Dataset 4 Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE
Fiber	1.700	2.209	0.857	2.893
Protein	8.330	8.472	1.437	3.229
Fats	22.094	22.080	0.920	1.849
Energy	281.312	288.811	20.640	53.468
Carbs	13.942	15.531	3.523	9.954

Statistical analysis result summarised in Table 3 and Table 4 indicated that randomising the order of ingredients significantly improved all measured statistical metrics, including MAE and RMSE, across all macronutrients. Moreover, all mean estimates closely approached the actual mean values. These improvements suggest that the sequencing of input ingredients influences the accuracy of the nutritional information outputs produced by the LLM.

Although this investigation was situated within the context of nutrition science, the implications might extend to the general application of LLMs. Further research is warranted to ascertain whether input sequencing similarly affects output accuracy in other domains. The findings emphasise the importance for researchers in nutrition science to incorporate input randomisation in their experimental designs, to mitigate potential biases in their results.

4.1.1.2 GPT-4-Turbo vs GPT-3.5-Turbo

In this comparative analysis of GPT-3.5-Turbo (dataset 4) and GPT-4-Turbo (dataset 3), the models were evaluated based on their ability to estimate the macronutrient and fibre content of 150 different food items. The evaluation metrics used include MAE and the RMSE, which are standard for assessing the accuracy of predictions in such contexts.

Table 5: Assessment of GPT-4-Turbo Nutrition Information Estimations Using Consistent Input Sequencing and Zero Temperature on Dataset 3 Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE
Fiber	1.700	2.269	0.714	2.383
Protein	8.330	8.794	1.592	3.497
Fats	22.094	21.960	0.711	1.464
Energy	281.312	291.444	22.272	59.020
Carbs	13.942	16.219	3.922	11.185

Considering the result in Table 5, GPT-4-Turbo demonstrated a strong ability in estimating the content of fibre and fats, showing a generally higher accuracy than GPT-3.5-Turbo with similar settings. This suggests that advancements in the model’s architecture may have improved its capability to handle these specific nutrients, perhaps due to better feature extraction or more nuanced understanding of textual inputs related to dietary content. In contrast, according to Table 4, GPT-3.5-Turbo exhibited a superior performance in estimating proteins, energy, and carbohydrates. This could imply that the earlier version of the model, while possibly less sophisticated, may be better tuned to certain types of data or it might benefit from its training regime which could have included more diverse or extensive dietary information. Additionally, the results confirms an overestimation of protein values by GPT-4-Turbo when compared to GPT-3.5-Turbo, which previously discussed by Hoang et al. (2023).

The findings illustrate a nuanced landscape where neither model consistently outperforms the other across all categories. This pattern suggests that each model has unique strengths and weaknesses in understanding and processing nutritional data. The variation in performance across different nutrients could be influenced by several factors includ-

ing the models’ training data diversity, the specific model architecture, and possibly the format and representation of the input data used for making estimations.

Furthermore, the comparison reveals important insights about the practical application of language models in dietary assessments. It highlights the potential for these models to assist in nutritional science but also underscores the need for careful consideration of each model’s capabilities and limitations. For instance, while GPT-4-Turbo shows promise in certain areas, its performance in energy estimation is notably weaker compared to GPT-3.5-Turbo, which could impact its utility in applications where energy content is a critical factor.

The literature review primarily discusses the enhanced reasoning capabilities of GPT-4-Turbo, noting its improvement over GPT-3.5-Turbo. These enhancements have enabled GPT-4-Turbo to significantly outperform its predecessor in complex reasoning benchmarks, such as the USMLE. However, the results of this study indicate that despite these advancements, GPT-4-Turbo still shows almost the same level of nutrition information accuracy in its responses. This discrepancy suggests that while GPT-4-Turbo’s enhanced reasoning abilities contribute to its superior performance in sophisticated cognitive tasks, these improvements have not been paralleled in tasks related to nutritional estimation. This identifies a critical area for potential enhancement in future iterations of the model.

Overall, the study underlines the importance of choosing the appropriate model based on the specific nutrient of interest. It also points to the potential benefits of developing hybrid approaches or ensemble methods that combine the strengths of both models to achieve more reliable and accurate predictions across a broader range of nutrients. This would not only enhance the models’ practical utility in nutritional science but could also pave the way for more personalised dietary recommendations and advanced food analysis techniques.

4.1.1.3 Model’s Temperature

Temperature is a crucial parameter that controls the variability of responses from LLMs, such as GPT. Typically, in tasks requiring high accuracy and replicability, researchers opt to set the temperature parameter to zero, minimising randomness to achieve consistent results. This study pioneers the investigation of the impact of temperature settings on the accuracy of LLM responses specifically in the field of nutrition science, a novel application

area for these models.

Comparing the data presented in the Table 6 against Table 3 illustrate a clear trend: increasing the temperature leads to a increasing in the precision of nutrition information estimated by the GPT model. Specifically, there is a noticeable, albeit slight, reduction in all error metrics as the temperature rises. However, the statistical analysis indicates that these differences, while consistent, are not significant. This suggests a robustness in the model’s ability to generate accurate nutrition information across a range of temperature settings.

Table 6: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Using Shuffled Input Sequencing and Temperature=0.7 on Dataset 5 Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE
Fiber	1.700	2.069	0.611	1.922
Protein	8.330	8.347	1.216	2.692
Fats	22.094	21.994	0.845	1.720
Energy	281.312	286.397	17.466	45.070
Carbs	13.942	15.178	2.916	8.274

The insensitivity of error metrics to temperature variations could imply that the underlying model has learned a sufficiently robust representation of nutrition-related data, such that its performance is stable or even better when introducing more randomness through higher temperature settings. This stability is crucial for practical applications where a balance between engaging, varied responses and factual accuracy is necessary.

These findings are instrumental for the development of AI-driven applications in nutrition, such as autonomous dietitian platforms. They suggest that developers can afford to use higher temperatures to achieve more natural, varied interactions without significant loss of information accuracy. This balance is particularly vital in user-facing applications like chatbots, where engagement and user satisfaction are key.

Our analysis reveals that among variables such as temperature, model type, and input sequence, the sequence of inputs exerts the greatest influence on the accuracy of nutritional information generated by GPT models. Notably, advanced models like

GPT-4-Turbo exhibit performance comparable to their predecessors, such as GPT-3.5-Turbo. Alterations in the model’s randomness, achieved by adjusting the temperature parameter, result in outputs that remain consistent and robust. However, modifications in the order of ingredients inputted significantly skew the results comparing to other variables. These findings underscore the importance of further research in the area of prompt engineering, especially for medical applications, rather than merely expanding model size. Additionally, techniques such as prompt engineering are significantly more cost-effective than model expansion, while potentially having a greater impact on output quality.

4.1.2 A Comprehensive Analysis of Nutritional Prediction Accuracy Across Macronutrients

In an extensive evaluation of the capabilities of the GPT-3.5-Turbo model to accurately predict the nutritional content of a diverse selection of 150 food items, a detailed analysis reveals significant insights into the model’s performance across various macronutrients and fibre. This analysis employed the SMAPE as a metric to quantify the relative errors associated with each nutrient type, shedding light on both the strengths and weaknesses of the model’s predictive accuracy. Additionally, this study discussed MAE and RMSE within each macronutrient category for a deeper analysis.

4.1.2.1 Energy Estimations: High Precision with Minor Outliers

The analysis of statistical metrics in Table 7 highlights that GPT-3.5-Turbo is most proficient in estimating the energy content of foods and corroborate the Hoang et al. (2023) findings. With a SMAPE of 10.783%, these predictions are notably accurate. The mean predicted values deviate minimally from the actual values, with an average discrepancy of merely 5 units. This precision suggests a robust model capability in energy prediction, likely due to the model’s effective processing of input data relevant to caloric content.

Table 7: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	1.700	2.024	0.637	2.072	16.530
Protein	8.330	8.354	1.263	2.813	17.668
Fats	22.094	21.996	0.874	1.759	27.075
Energy	281.312	286.373	18.213	46.628	10.793
Carbs	13.942	15.213	3.022	8.635	23.453

However, while the MAE is relatively low, indicating small average errors, the RMSE is nearly three times greater. This discrepancy between MAE and RMSE points to the presence of significant outliers where the model’s predictions are substantially off the mark—some by as much as 250 kcal. In Figure 9, the residual plots for energy values, however, do not display any discernible bias, implying that these errors are sporadic rather than systematic. This observation is supported by Figure 8, which indicates that the majority of energy predictions closely align with actual values, affirming the model’s reliability in this aspect.

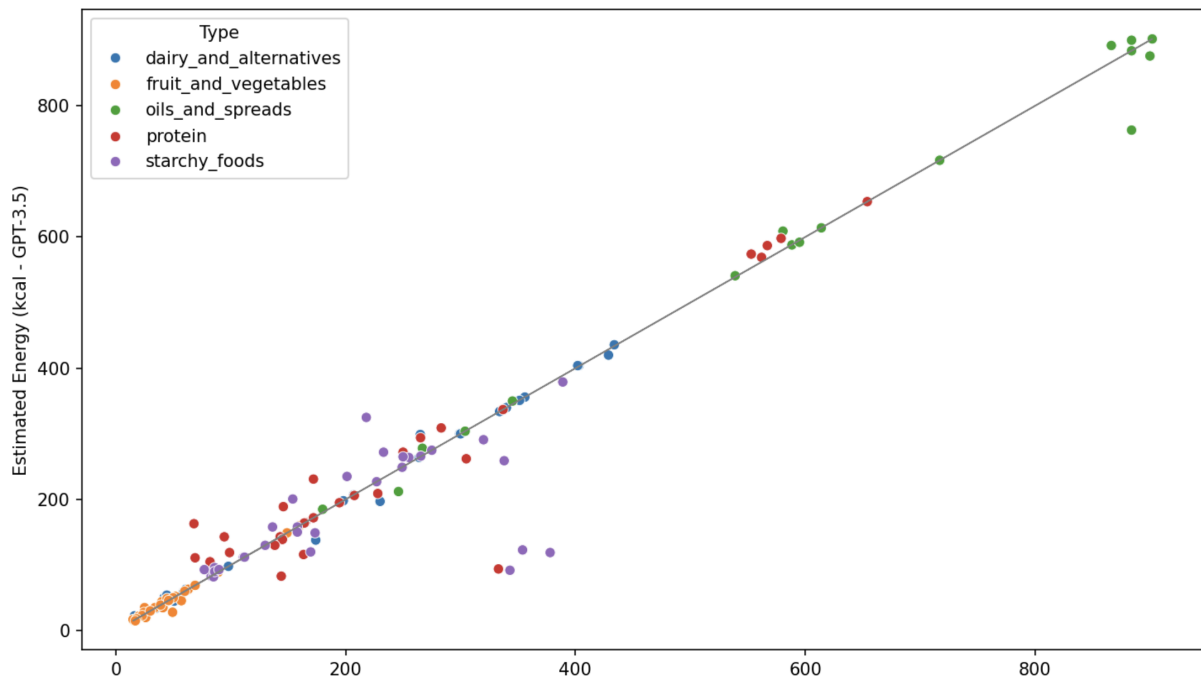


Figure 8: Comparison of energy values from the NutritionIX (USDA) and GPT-3.5-Turbo.

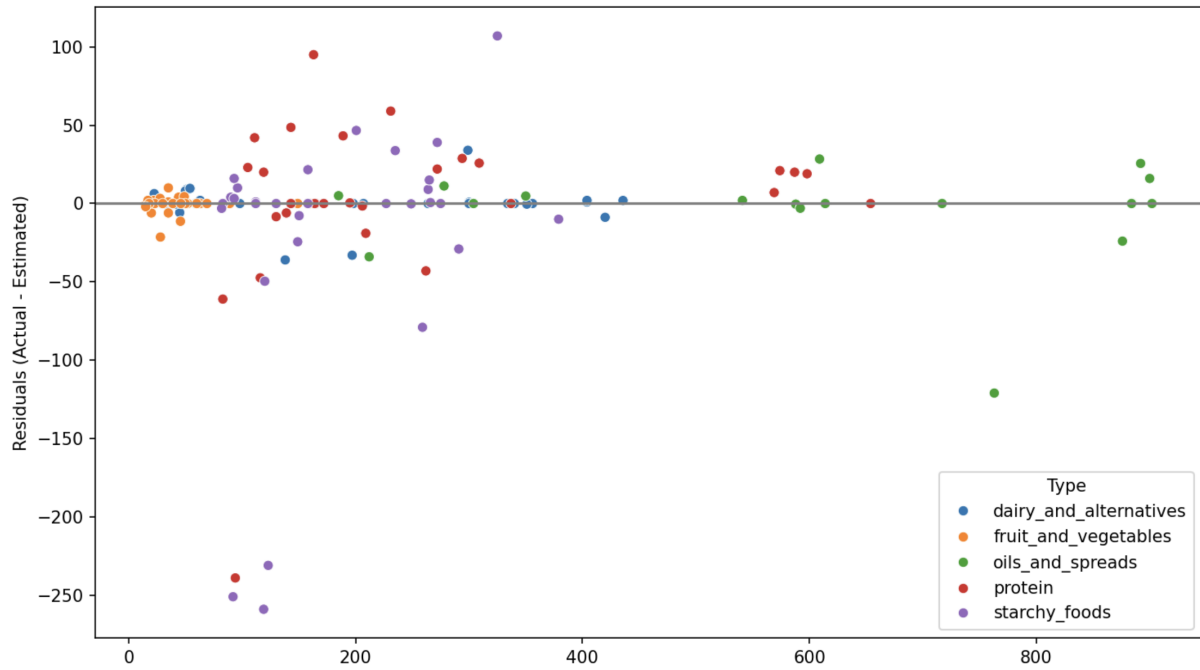


Figure 9: Residual plot of energy values from the NutritionIX (USDA) vs GPT-3.5-Turbo.

4.1.2.2 Fibre Estimations: Tendency to Overestimate

Moving on to fibre, based on Table7, the model achieves a SMAPE of 16.530, positioning it as the second most accurate prediction after energy. Despite this relative accuracy, the model exhibits a consistent tendency to overestimate fibre content. This overestimation is highlighted by the substantial MAE, which amounts to nearly 30% of actual fibre values. Such a significant MAE, coupled with a relatively moderate SMAPE, suggests that certain outlier predictions, which deviate greatly from actual values, disproportionately influence the error metrics which can be confirm according to Figure 10. As fibre content increases, these prediction errors tend to magnify, though the residual plots in Figure 11 show no particular bias, maintaining a normal distribution of residuals.

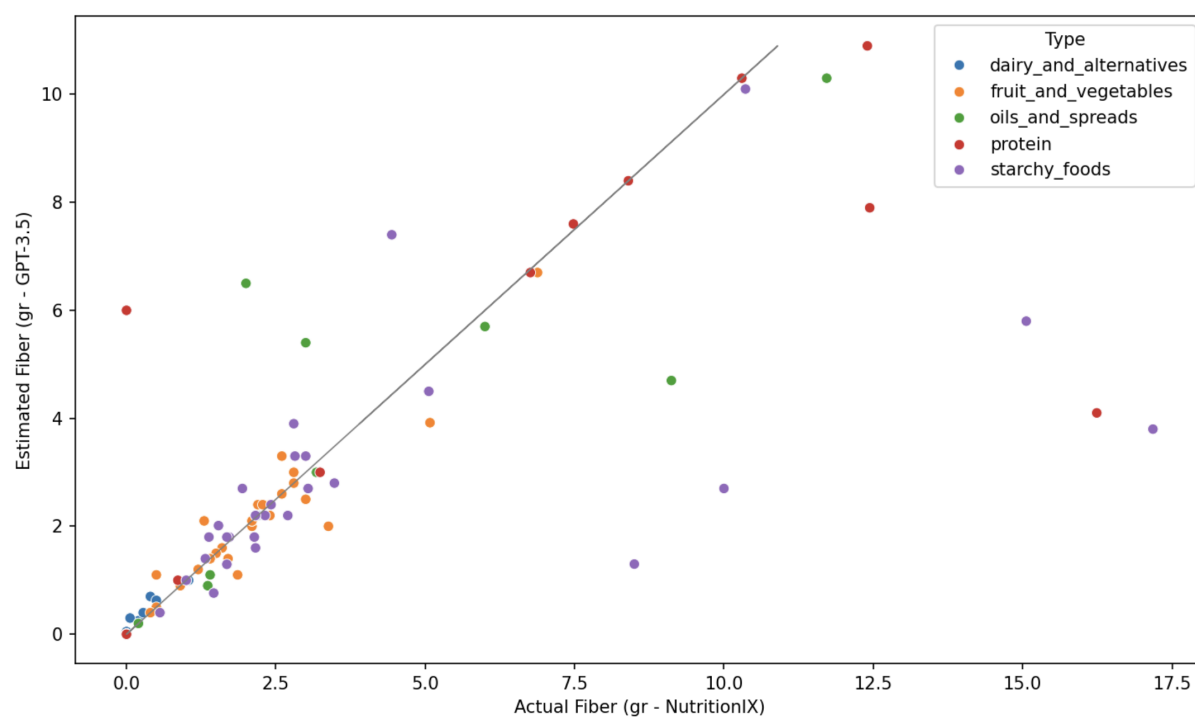


Figure 10: Comparison of fibre values from the NutritionIX (USDA) and GPT-3.5-Turbo.

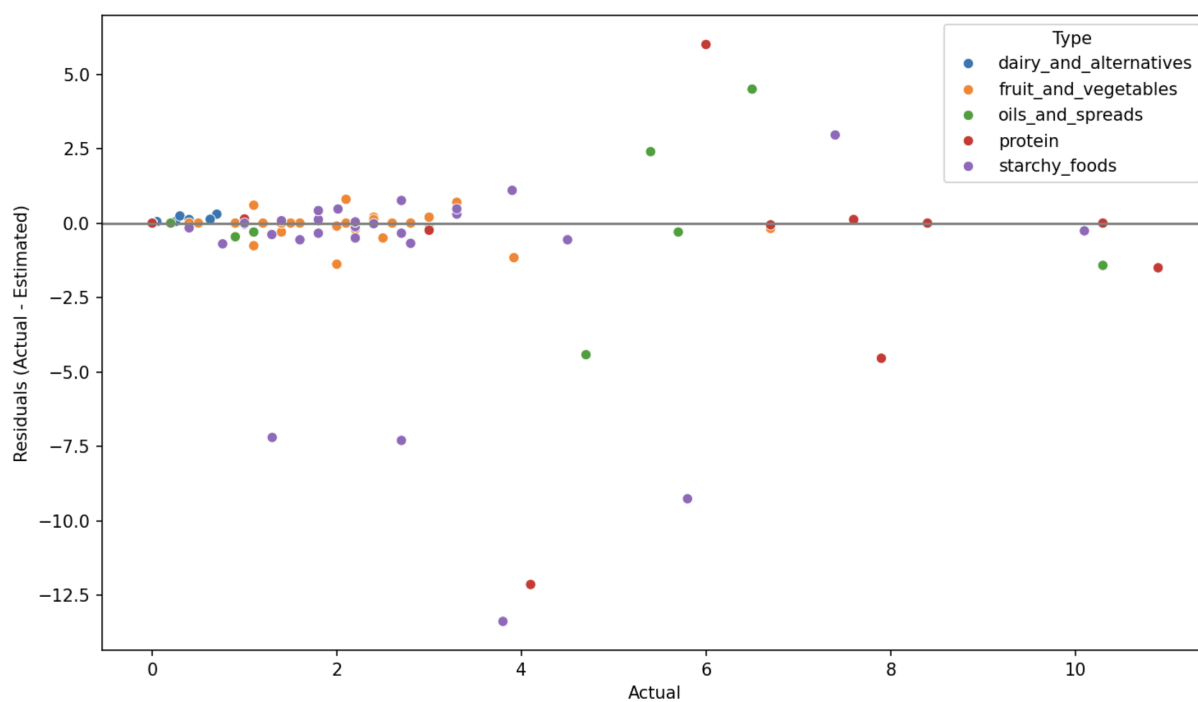


Figure 11: Residual plot of fibre values from the NutritionIX (USDA) vs GPT-3.5-Turbo.

4.1.2.3 Protein Estimations: Uniform Accuracy

Protein predictions by GPT-3.5-Turbo also display a SMAPE comparable to that of fibre. The closeness of mean actuals to mean estimates suggests a general accuracy in the model’s protein estimations. However, the relatively high MAE and RMSE compared to the mean actual values indicate that, although the model does not systematically overestimate or underestimate protein content, the spread of errors around the mean is significant. Figure 12 and Figure 13 illustrate that the model’s accuracy in predicting protein content is consistent across foods with varying protein levels, with most significant outliers occurring in items with mid-range protein.

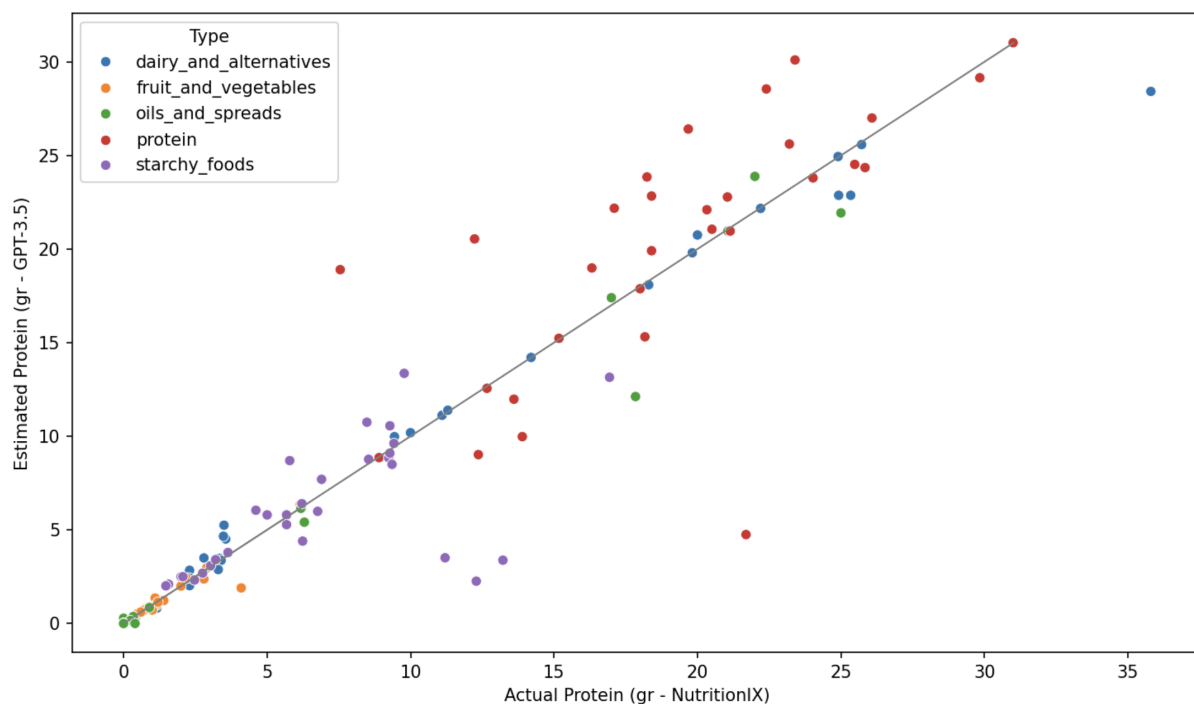


Figure 12: Comparison of protein values from the NutritionIX (USDA) and GPT-3.5-Turbo.

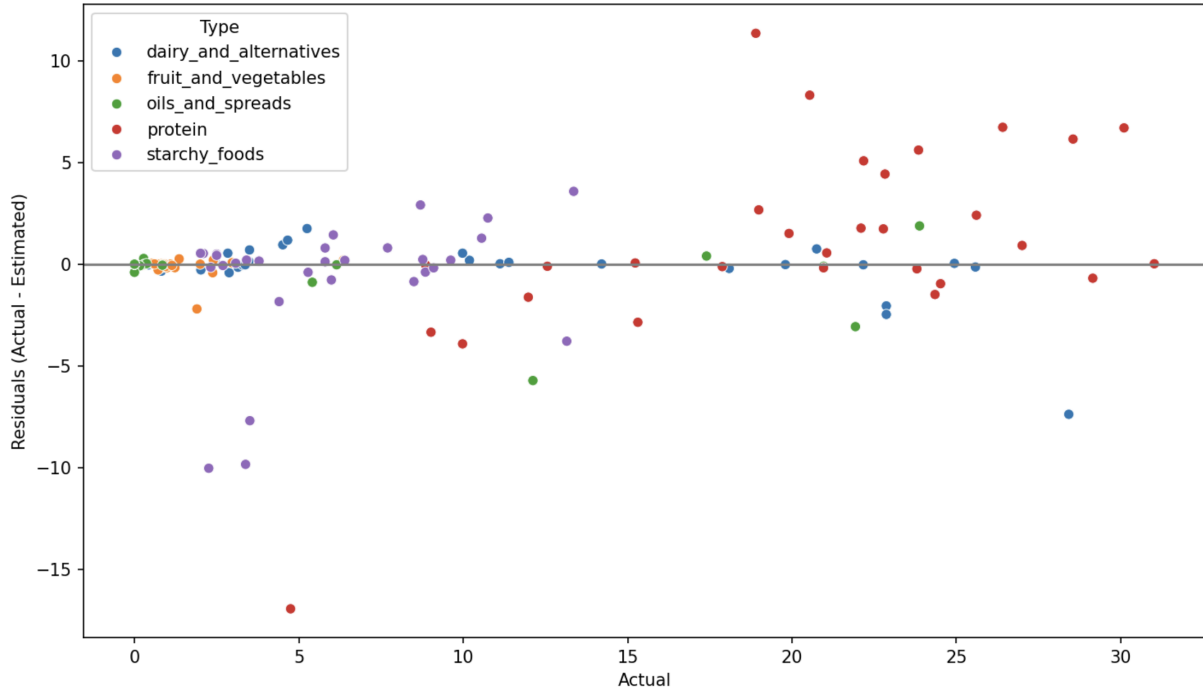


Figure 13: Residual plot of protein values from the NutritionIX (USDA) vs GPT-3.5-Turbo.

4.1.2.4 Carbohydrate Estimations: Overestimation Issues

Carbohydrates present a greater challenge for GPT-3.5-Turbo, with a SMAPE of 23.453 regarding Table 7. The model tends to overestimate carbohydrate content, as evidenced by a MAE of around 3 and an RMSE that is more than 2.5 times the MAE. This pattern suggests that errors in carbohydrate predictions are not just frequent but also pronounced, particularly in items with higher carbohydrate content. Figure 14 and Figure 15 confirm that the model estimations significantly vary with actual values in several instances from low-range to high-range, pointing to a need for recalibration considerations in this nutrient category.

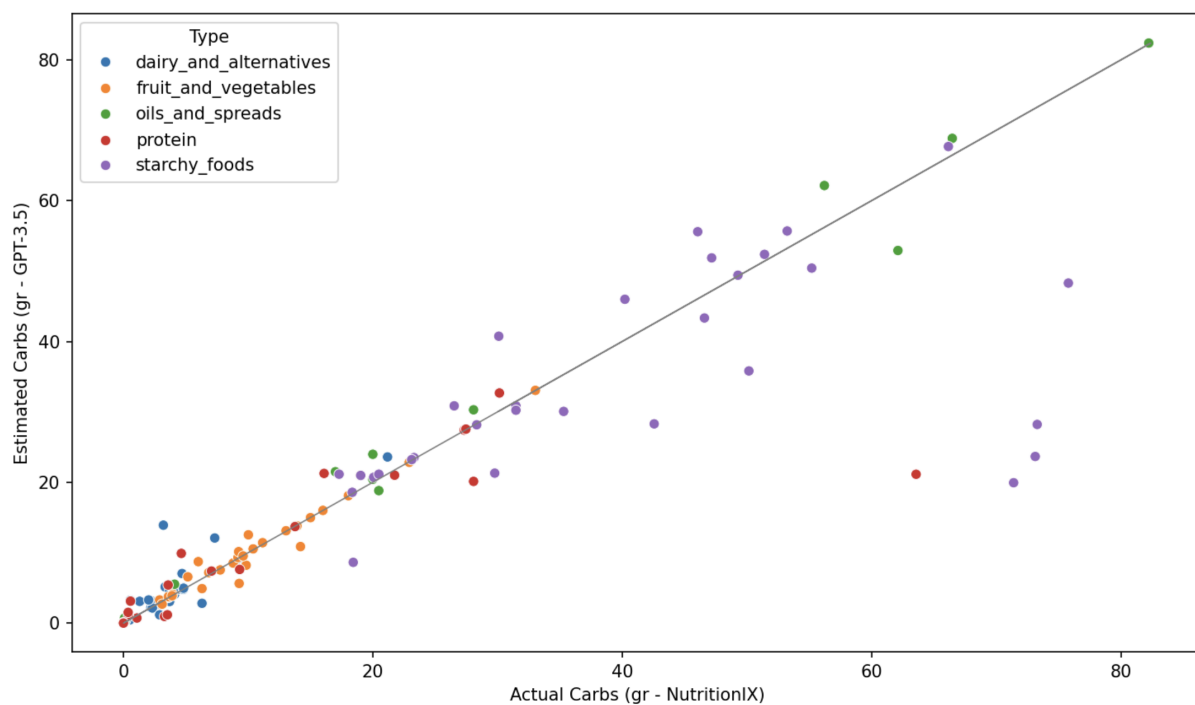


Figure 14: Comparison of carbohydrate values from the NutritionIX (USDA) and GPT-3.5-Turbo.

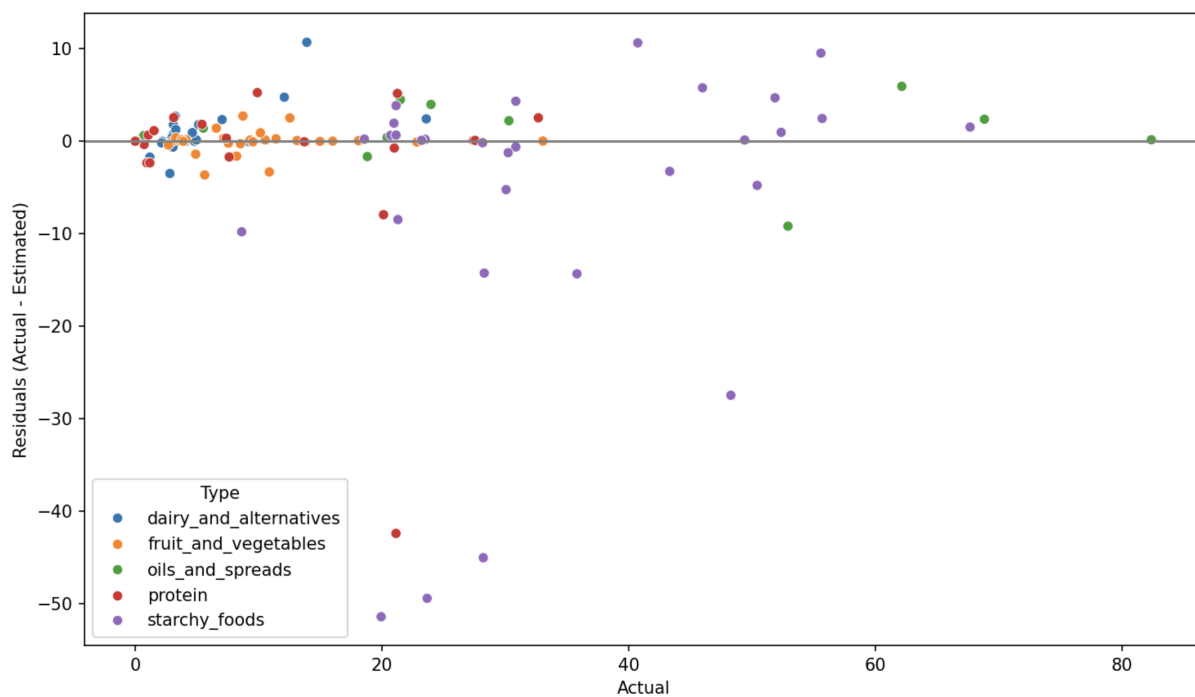


Figure 15: Residual plot of carbohydrate values from the NutritionIX (USDA) vs GPT-3.5-Turbo.

4.1.2.5 Fat Estimations: Complex Error Dynamics

Estimations of fat content reveal the most complex error dynamics, with the highest SMAPE of around 27 according to Table 7. Despite this high error rate, the mean actuals and mean estimates align perfectly, and both MAE and RMSE are unexpectedly low. This paradox indicates that the major prediction errors predominantly occur in items with low fat content, where the model struggles to estimate accurately. Analysis from Figure 16 shows that predictions are highly accurate for foods with mid-range and high-range fat content and reasonably so for those with high fat content. Conversely, Figure 17 reveals high residuals primarily in low-fat items, suggesting specific difficulties in modelling foods with minimal fat content.

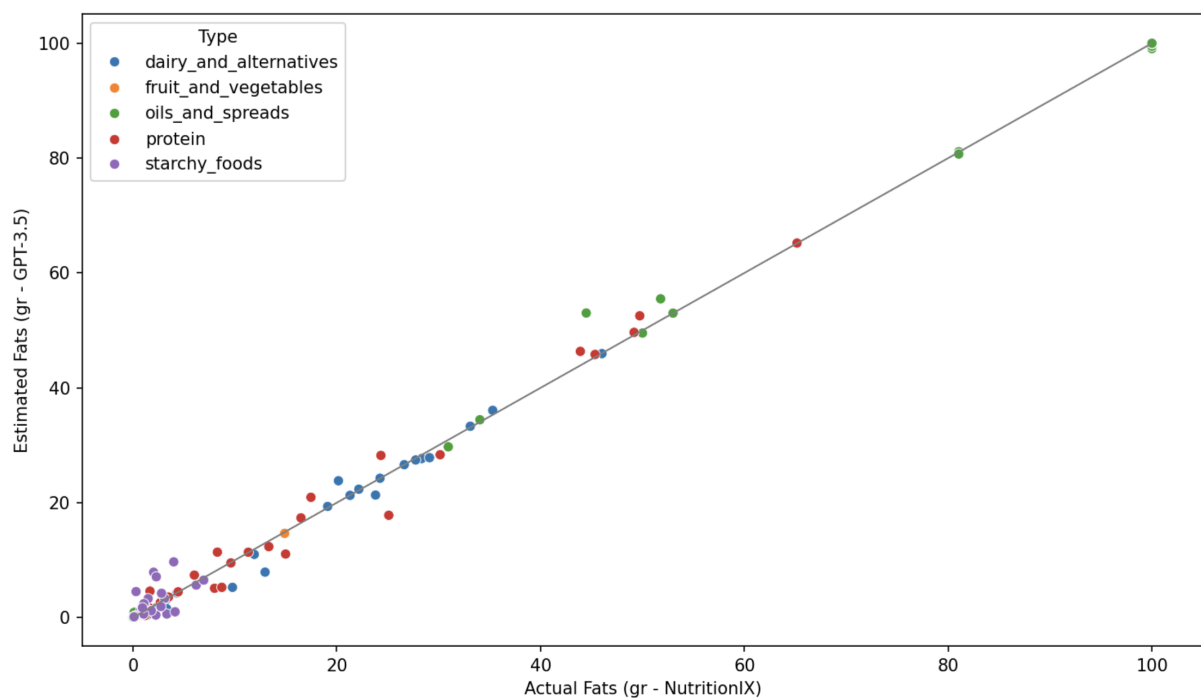


Figure 16: Comparison of fat values from the NutritionIX (USDA) and GPT-3.5-Turbo.

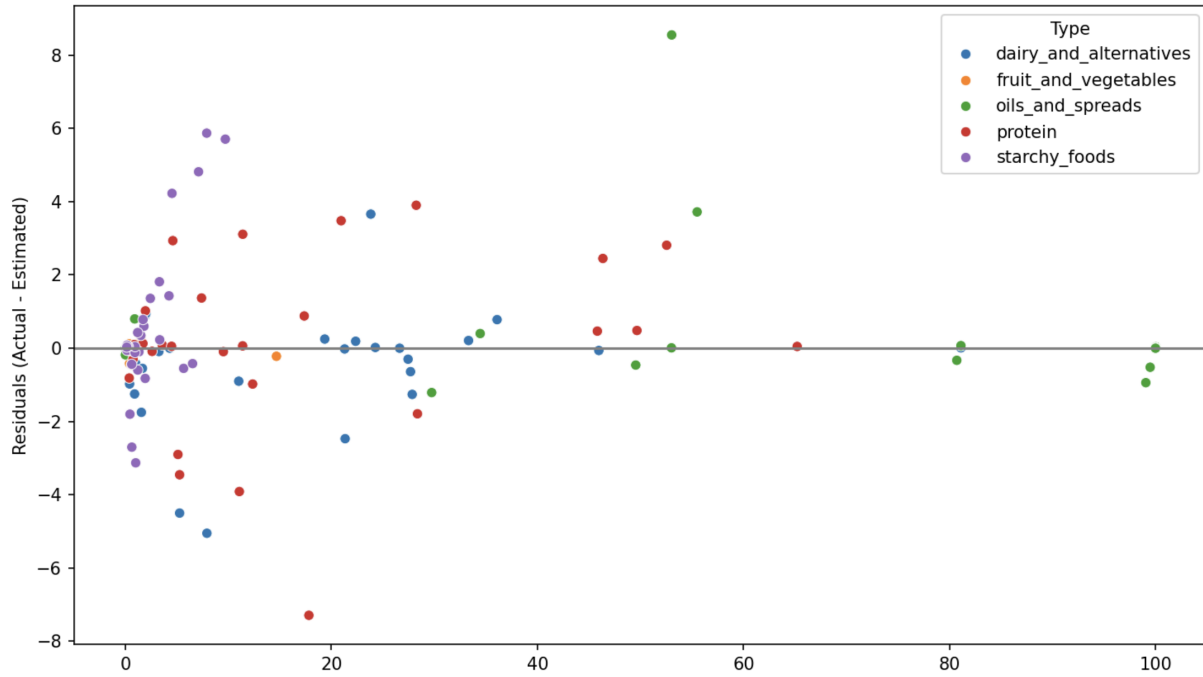


Figure 17: Residual plot of fat values from the NutritionIX (USDA) vs GPT-3.5-Turbo.

4.1.2.6 Implications for Practical Application

The current performance of GPT-3.5-Turbo, while adequate for providing general nutritional overviews or for use in non-critical applications, may not meet the needs of scenarios requiring precise dietary planning or clinical accuracy. Enhancing the model's accuracy and reliability could significantly extend its utility into more demanding areas such as personalised nutrition planning or dietary management in clinical settings.

4.1.2.7 Strategic Enhancements

Given these insights, there is a clear necessity to improve model performance, especially in accurately predicting low-value but nutritionally critical components like fibre and carbohydrates. Adopting advanced techniques such as RAG could potentially address and mitigate the observed biases and inaccuracies. Such enhancements would not only increase the utility of machine learning models in nutritional science but also pave the way for more reliable and robust dietary assessment tools, thereby benefiting the health and wellness industries. This strategic focus on enhancing precision and reliability in nutritional predictions underscores the ongoing evolution of AI capabilities in specialised domains.

4.1.3 Comparative Analysis of Macronutrient Estimation Across Food Categories by GPT-3.5-Turbo

In this section, a detailed analysis is conducted on the precision of nutritional information estimates provided by GPT-3.5. The methodology previously outlined is followed for this examination. Foods and ingredients are categorised into five distinct groups in alignment with NHS guidelines. It is observed that the accuracy of the model varies notably among these categories. The SMAPE is utilised as a metric to compare these variations, with energy value predictions serving as a benchmark for assessing accuracy across different food categories due to their comparative reliability. Each category encompasses five attributes, including macronutrients and fibre, making the comparison of SMAPE across the groups challenging. Consequently, an average of the SMAPE values for each category is calculated, and the performance of the categories is then ranked based on these average SMAPE scores.

4.1.3.1 Fruit and Vegetables

In this analysis, GPT-3.5-Turbo exhibited the greatest accuracy in estimating the nutritional information of fruits and vegetables, achieving an average SMAPE of 11.52. As illustrated in Table 8, this accuracy surpasses estimations for other food categories, In contrast, as indicated in Table 7, the average SMAPE for all foods and ingredients stood at 19.10. Consequently, the estimates for the fruits and vegetables category were found to be 40% more accurate on average compared to other categories.

Table 8: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients for Fruit and Vegetable Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	2.081	2.146	0.240	0.441	11.624
Protein	1.368	1.465	0.167	0.425	9.904
Fats	0.770	0.781	0.066	0.114	18.926
Energy	46.911	47.273	2.807	5.533	7.923
Carbs	10.297	10.293	0.751	1.295	9.241

Figure 8 highlights that energy values are estimated with the highest precision, re-

flected in the lowest SMAPE values. Conversely, fat estimations are the least accurate, with a SMAPE of 18.92. Additionally, the mean estimates and actual nutritional values across all categories show a high degree of similarity, as evidenced by small MAE and RMSE values. This consistency suggests that the model’s estimations for this category are not only accurate but also exhibit minimal variability, indicating few outliers in the predictions.

4.1.3.2 Oils and Spreads

The analysis of the GPT-3.5-Turbo model’s performance in estimating the nutritional content of oils and spreads highlights its strengths and identifies areas for improvement. According to Table 9, the model demonstrates exceptional accuracy in predicting the energy content of these products, achieving the lowest SMAPE among all categories at 1.70. This low SMAPE underscores the model’s effectiveness in caloric estimation. The MAE and RMSE are around 7 and 25, respectively, which, in comparison to the mean estimates of approximately 720 calories, are relatively minor. Most deviations occur with high-calorie items, as depicted in Figure 9. Furthermore, the mean estimates and actual mean values differ by only 3 units, a discrepancy that is highly significant, particularly for oils and spreads where the primary concern revolves around caloric values. Therefore, the model’s responses in this area can be considered reliable.

Table 9: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients for Oils and Spreads Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	1.260	1.266	0.466	1.265	10.518
Protein	3.660	3.920	0.432	1.250	24.404
Fats	74.554	74.225	0.577	1.739	15.308
Energy	717.744	720.713	9.191	24.683	1.698
Carbs	12.922	12.563	1.084	2.393	10.928

For carbohydrates and fibre values, the model’s precision based on SMAPE is nearly identical. However, the relatively high MAE for fibre suggests that most misestimations occur in items with high fibre content within this category, as shown in Figure 10. On the other hand, the least accurate macronutrient prediction is for protein, with an RMSE

of 24.40. This increased relative error is likely due to the typically low baseline levels of protein in oils and spreads, where even small absolute errors result in larger percentage errors.

4.1.3.3 Dairy and Alternatives

With a SMAPE of 18.56, the GPT-3.5-Turbo model demonstrates moderate accuracy in estimating the nutritional content of dairy products and alternatives when compared to the overall population. Notably, regarding Table 10 energy value estimations in this category are even more accurate than those for fruits and vegetables, exhibiting a lower SMAPE. Despite occasional deviations from actual caloric values, the model’s overall capability to estimate energy content remains robust. This precision is particularly vital for managing caloric intake within dietary regimes, where accurate energy estimation can profoundly influence nutritional balance.

Table 10: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients for Dairy and Alternatives Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	0.184	0.156	0.031	0.079	15.747
Protein	10.205	10.425	0.691	1.553	9.706
Fats	17.143	17.581	0.919	1.599	23.483
Energy	212.445	214.827	7.364	15.985	7.187
Carbs	4.800	3.980	1.236	2.460	36.677

The model also displays relatively high accuracy in protein estimation, with a SMAPE of 9.71. The MAE relative to mean estimates for both energy and protein is considerably low, underscoring the model’s reliability for dietary planning. This accuracy is especially critical for individuals who rely on dairy alternatives as their primary protein sources.

However, the model faces significant challenges in accurately estimating fats and carbohydrates. The SMAPE for fats is 23.483, while for carbohydrates, it escalates dramatically to 36.677. These elevated SMAPE values suggest substantial relative errors, with the carbohydrate estimations showing an approximate 20% difference between the mean estimates and actual values. Additionally, Figure 15 highlights some outliers in

low-carbohydrate items within this category, further emphasising the discrepancies in carbohydrate estimations.

4.1.3.4 Protein

In the analysis of the GPT-3.5-Turbo model’s accuracy within the protein category, the average SMAPE for macronutrients is recorded at approximately 22.13 in Table 11. This indicates notably lower estimation precision compared to other categories, where SMAPE values for energy were consistently below 10. In this category, however, the SMAPE for energy rises to around 18. Furthermore, as depicted in Figure 8, most misestimations occur in the lower-to-mid range of energy values, suggesting a particular vulnerability in accurately predicting energy levels in items with moderate caloric content.

Table 11: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients for High Protein Foods Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	2.197	2.604	0.825	2.623	13.404
Protein	20.339	19.222	3.283	5.011	19.418
Fats	15.612	15.571	1.518	2.292	22.714
Energy	247.200	245.553	30.007	54.180	18.065
Carbs	7.461	8.735	2.589	8.065	37.068

Significantly, the accuracy of protein estimations, which are central to this category, presents a normal performance with a SMAPE of 19.42. The comparison between mean estimates and actuals reveals a general tendency of the model to underestimate protein values, with MAE and RMSE being relatively high. These deviations highlight the substantial inconsistency of the estimations from the actual values. The red points in Figure 12 further corroborate these findings, emphasising the inaccuracy in protein estimations. Given that items in this category contribute substantially to daily protein intake, discrepancies from reference values could potentially lead to protein deficiencies for users, making accuracy here crucial.

Additionally, carbohydrate estimations within this category significantly diverge from actual values, exhibiting the poorest performance among all macronutrients with a SMAPE of 37.07. The RMSE is notably three times higher than the MAE and exceeds the mean

actual values, indicating a prevalence of outliers in the predictions. Such substantial errors underscore the model’s unreliability in predicting carbohydrate content within the protein category, reflecting a critical area for improvement to enhance the model’s utility in nutritional assessments.

4.1.3.5 Starchy

The evaluation of the GPT-3.5-Turbo model in estimating the nutritional content of starchy foods uncovers significant discrepancies across various nutrients, with an overall average SMAPE of 30.73. This analysis reveals the model’s varying degrees of accuracy for fibre, protein, fats, energy, and carbohydrates, indicating areas where improvements are critically needed.

Based on result in Table 12, the model particularly struggles with fat content estimations, presenting the highest SMAPE at 54.947. This large error margin suggests that the model’s algorithms may not adequately capture the variability of fats in starchy foods, which is concerning given the importance of accurate fat content estimation in nutritional planning. Fibre predictions also demonstrate notable inaccuracies, with a SMAPE of 31.357, further reflecting the model’s limitations in accurately gauging dietary fibre, a key component for digestive health and metabolic regulation.

Table 12: Assessment of GPT-3.5-Turbo Nutrition Information Estimations Across Macronutrients for Starchy Foods Compared to Actual Data.

	Mean Actuals	Mean Estimates	MAE	RMSE	SMAPE
Fiber	2.779	3.946	1.624	3.575	31.357
Protein	6.078	6.738	1.740	3.208	24.907
Fats	2.392	1.822	1.291	2.150	54.947
Energy	182.261	203.500	41.695	83.904	19.094
Carbs	34.229	40.494	9.447	17.156	23.353

In terms of protein, the model estimates a slightly higher content than observed, with a SMAPE of 24.907. While the errors here are less severe than for fats, the moderate MAE of 1.740 and RMSE of 3.208 indicate that the model still faces challenges in consistently predicting protein levels accurately. These protein estimation errors could mislead dietary assessments, particularly for diets where precise protein intake is necessary.

Carbohydrate estimations also fall short of accuracy, with a SMAPE of 23.353. Given that carbohydrates are a major energy source in starchy foods, the model's overestimation, indicated by an MAE of 9.447 and RMSE of 17.156, could have significant implications for dietary management, especially in carbohydrate-sensitive conditions such as diabetes.

Energy content predictions show relatively lower inaccuracies compared to other nutrients, with a SMAPE of 19.094. However, the high MAE of 41.695 and RMSE of 83.904 highlight the presence of substantial outliers in the model's energy estimations. While the error percentage is lower, the magnitude of these errors is considerable and could affect caloric management strategies in dietary planning.

4.1.3.6 Discussion

The comparative analysis across different food categories using the GPT-3.5-Turbo model demonstrates a varied landscape of accuracy in estimating nutritional information. The model exhibits the highest precision in estimating the nutritional content of fruits and vegetables and shows reliable caloric estimations for oils and spreads. However, it encounters significant challenges with certain macronutrients, especially fats and carbohydrates in starchy foods and dairy alternatives. These findings underscore the necessity for targeted improvements in the model's algorithms, particularly for categories where the estimation errors could significantly impact dietary management and health outcomes.

Moreover, the consistent use of SMAPE as a comparative metric reveals not only the areas of strength but also the critical weaknesses of the model, particularly in categories that are essential for balanced nutrition. As the technology continues to evolve, focusing on reducing errors in macronutrient estimations will be crucial for enhancing the model's utility in nutritional science and dietary planning. This analysis not only provides a foundation for further refinement of AI capabilities in nutritional assessments but also highlights the potential of AI tools like GPT-3.5-Turbo in contributing to more personalised and accurate dietary recommendations.

4.2 Enhancing Nutrition Accuracy Using Retrieval-Augmented Generation and NutritionIX

This study yielded diverse results, encompassing the success rate of API calls to the NutritionIX, the accuracy of nutritional information obtained from third-party databases, and the practicality of the employed methodology in real-world applications. The methodology of this research ensured that the food items retrieved from the dataset accurately represent those found in real-world diet plans. Consequently, the analysis of these results can be generalised to real-world scenarios where autonomous systems like GPT are used for generating diets.

4.2.1 API Success rate

This experiment employed NutritionIX as an external database to supplement real nutrition information for this study. The API exhibited an exemplary success rate, achieving a 100% response rate for all calls made, indicating that each request to the API was successfully fulfilled. This level of reliability underscores the API's robustness in providing consistent access to nutrient data.

Despite this high success rate, there are nuances in the quality of data retrieved that merit further exploration. Notably, the API did encounter instances where specific nutritional values, such as fibre content, were not available (returned as null). This highlights areas where the database may require enhancements to ensure comprehensive nutrient information coverage.

Overall, the flawless API call success rate holds significant implications for the future integration of such external databases with Large Language Models. This integration has the potential to enrich the models' capabilities by providing extensive and reliable nutrition data, thereby expanding their applicability in dietary and health-related research. The forthcoming sections will further analyse the accuracy of the data retrieved from these API calls, which is crucial for validating the utility of integrating NutritionIX API with advanced computational models.

4.2.2 Accuracy of Nutritional Information obtained from NutritionIX in Practice

In the proposed strategy to improve the accuracy of nutrition information in the model, although the data retrieval success rate was 100%, not all data precisely matched the corresponding food items in the diet meals. To evaluate the model’s practical efficiency, this study classified the API responses into three categories: identical, variant, and misinterpreted. The ‘identical’ category comprises items whose names exactly match those in the third-party database as listed in the diet plans. The ‘variant’ category includes items that are essentially the same but differ in spelling, such as ‘lactose-free milk’ versus ‘lactose free milk’. The ‘misinterpreted’ category contains items where the API name is related but not identical to the diet plan name, such as ‘tofu’ compared to ‘pan fried tofu’.

This analysis involved 20 synthesised meal plans tailored to various user profiles with distinct dietary restrictions, encompassing 162 items in total. The results, illustrated in Figure 18, indicate that the proposed strategy successfully augmented factual nutrition information to the GPT responses for over 90% of the items. However, not all items were precisely matched: 71.6% were identical, 18.5% were variants, and 9.9% were misinterpreted. Misinterpretations may stem from the NLP limitations of the NutritionIX API or ambiguous item descriptions in the GPT-generated diet plans. For instance, discrepancies arose when GPT specified cooking methods, such as frying, which the NutritionIX API failed to recognise.

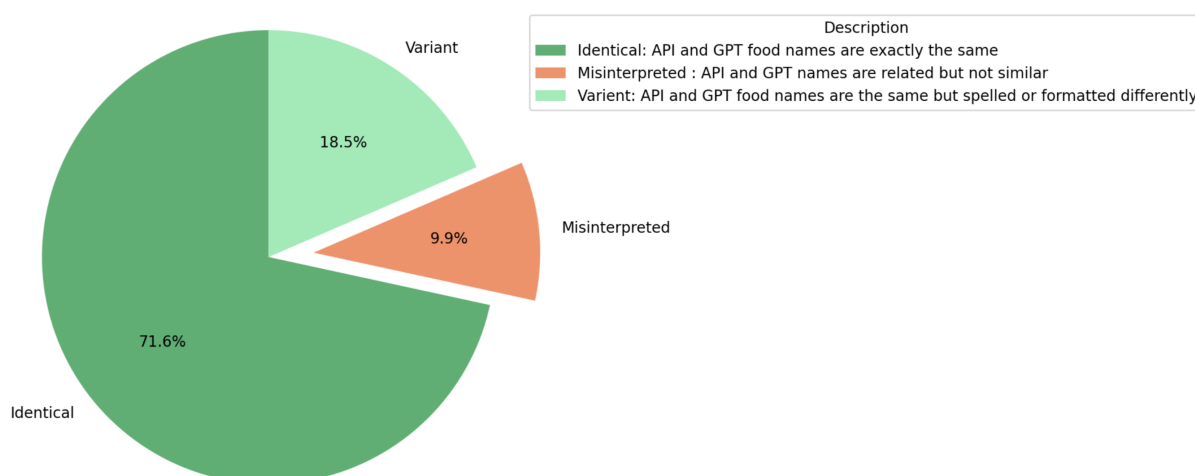


Figure 18: Comparison of fibre values from the NutritionIX (USDA) and GPT-3.5-Turbo.

According to prior experiment in this research and the findings of Hoang et al. (2023) on the same field, the nutrition accuracy of responses generated by GPT is relatively high as a baseline. By integrating the proposed strategy with the current capabilities of LLMs, their accuracy could potentially improve by up to 90%, marking a significant enhancement. This strategy offers promising applications for future research and could be seamlessly integrated into the data pipelines of autonomous diet plan generators, nutritionist chatbots, and other LLM-powered applications, thereby facilitating the workflow for human practitioners.

4.2.3 Discussion

The implications of these findings extend beyond mere data accuracy enhancement. This study underscores the potential of LLMs to revolutionise dietary planning and nutritional assessment by leveraging improved NLP techniques. By refining the interaction between LLMs and external nutritional databases, the accuracy and reliability of dietary information can be significantly enhanced, leading to better-informed dietary choices for individuals.

Furthermore, the ongoing challenges with API misinterpretations highlight an area ripe for future research. Enhancements in contextual understanding and the semantic analysis capabilities of LLMs could reduce the frequency of these misinterpretations. Research could explore more sophisticated models that incorporate a broader range of linguistic nuances and dietary contexts such as KB recommender systems, thereby improving the precision of dietary information provided to users.

Additionally, integrating these improvements into user-friendly applications can democratise access to personalised nutrition guidance, potentially reducing health disparities and supporting public health initiatives focused on diet-related chronic diseases. Ultimately, as LLMs continue to evolve, their integration into the health and wellness sectors appears both promising and transformative, setting a new standard for how dietary information is processed and utilised in healthcare and everyday life.

5 Discussion and Conclusion

This study commenced with a systematic review of current GPT models in nutrition science, assessing their performance in delivering nutritional values and proposing a strategy to enhance their accuracy. The methodology employed encompassed a broad range of experiments and settings, exploring the impact of model selection, configuration, and prompt engineering on output quality.

The study evaluated these models' abilities to estimate macronutrients in various foods, conducting a thorough analysis of their accuracy across different food groups. This investigation not only highlighted the general robustness of these models in estimating nutritional content but also underscored significant disparities in performance among food categories. Notably, models performed better with simpler items, such as fruits and vegetables, whereas they struggled with more structurally complex items like starchy foods, indicating a need for specialised improvements in these areas.

A significant contribution of this research is the proposed strategy utilising the Retrieval-Augmented Generation technique, which markedly improved model accuracy, reaching up to 90%. Future enhancements could include more advanced techniques or the integration of sophisticated external knowledge bases, potentially elevating model performance further.

Despite these advances, the reliability of GPT models for clinical applications in nutrition remains questionable, primarily due to the critical nature of precise nutritional information and its impact on health outcomes. Nonetheless, these models are invaluable as tools under the supervision of dietitians, assisting practitioners in formulating advice and diet plans. However, it is crucial that model outputs are vetted by professionals before being relayed to clients.

Previous studies, such as that by Denniss et al. (2023), have critiqued the accuracy of nutrition information available online and via social media. In contrast, this research finds GPT-generated nutrition information to be relatively accurate, suggesting that large language models like GPT could serve as more reliable alternatives for public inquiries than traditional online searches.

Further research is necessary, particularly in the domains of meal planning and allergen detection, where concerns about model reliability, performance, and the acceptance of

responses remain prevalent. Enhancing these aspects could lead to broader application and acceptance of these models in nutrition science.

Moreover, ethical considerations must be integrated into future research and applications to ensure the responsible deployment of GPT models in nutrition science. Transparency in model operations is critical; researchers and developers should clearly articulate how models generate their outputs and the limitations inherent within them. This includes providing understandable explanations that can be comprehended by both practitioners and clients, ensuring that all users are fully informed about the basis of the nutritional advice they are receiving.

Additionally, addressing potential biases in training data is crucial. GPT models, like other AI technologies, are susceptible to replicating or amplifying biases present in their training datasets. In the context of nutrition, this could manifest in skewed nutritional suggestions that may not be suitable for all populations. Efforts should be made to curate diverse and inclusive datasets that accurately reflect the wide range of dietary needs across different demographics, cultures, and individual health requirements.

Safeguarding user data is another significant ethical issue. Nutrition-related applications often involve sensitive personal information, including health data that users may input to receive customised dietary recommendations. It is imperative that robust data protection measures are in place to protect this information from unauthorised access or breaches. Clear, strict privacy policies and practices should be established and communicated to users, ensuring their data is handled securely and with respect.

In line with these ethical considerations, the development of ethical guidelines specifically tailored to the use of AI in nutritional science could further support the responsible use of these technologies. These guidelines could provide frameworks for addressing ethical challenges, conducting risk assessments, and implementing safeguards that protect both users and the integrity of the nutritional advice provided.

By proactively addressing these ethical concerns, the field can harness the potential of GPT models to provide valuable nutritional insights while ensuring they are used in a manner that is both ethical and beneficial to all stakeholders involved.

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A Prompt for Diet Generation that Adheres to NHS Guidelines for a Balanced Diet

Generate a 1-day personalized diet plan based on the following profile and dietary preferences. Ensure the diet plan aligns with the principles of a balanced diet, incorporating guidelines from the Eatwell Guide. Each day should include meals (breakfast, lunch, dinner) and two snacks, emphasizing a variety of foods from all the major food groups.

****Profile Details:****

- Age: [Insert Age]
- Gender: [Insert Gender]
- Weight: [Insert Weight]
- Height: [Insert Height]
- Activity Level: [Insert Activity Level (e.g., sedentary, lightly active, very active)]
- Dietary Restrictions: [Insert any allergies, intolerances, or preferences e.g., gluten-free, nut-free, vegetarian, vegan]
- Health Goals: [Insert specific health goals e.g., weight loss, muscle gain, maintenance, specific nutrient focus]

****Daily Nutritional Goals:****

- Calories: [Insert daily caloric goal]
- Carbohydrates: [Specify percentage of calories from carbohydrates]
- Protein: [Specify percentage of calories from protein]
- Fats: [Specify percentage of calories from fats]
- Fiber: [Specify amount in grams]
- Specific Nutrients to Focus on: [Insert specific vitamins, minerals, etc., based on personal health needs]

****Day-by-Day Meal Structure:****

- ****Day 1:****
- Breakfast:
- Lunch:
- Dinner:
- Snack 1:
- Snack 2:

Please ensure the plan:

1. Includes at least five portions of a variety of fruits and vegetables per day.
2. Uses whole grain or high fiber starchy carbohydrates as the basis of meals.
3. Incorporates a variety of protein sources, focusing on plant-based options and including dairy or fortified alternatives.
4. Limits the intake of saturated fats, sugars, and salt.
5. Recommends six to eight cups of fluids per day.
6. Addresses any specific dietary considerations noted in the profile (e.g., vegetarian, vegan needs).

Please provide the amount of each item in grams.

Once you created the meal plan and give a comprehensive considerations based on criteria and user profile, at the end, also provide the diet plan in the a structured output outline in the following format:

```
```json
{
 "days": [
 {
 "day": "{{day}}",
 "meals": [
 {
 "name": "{{mealName1}}",
 "foods": [
 {"food_name": "{{foodName1}}", "amount": {{amount1}}, "energy_per_100g": {{energy1}}, "fat": {{fat1}}, "fiber": {{fiber1}}, "protein": {{protein1}}, "carbohydrates": {{carbohydrates1}}},
 {"food_name": "{{foodName2}}", "amount": {{amount2}}, "energy_per_100g": {{energy2}}, "fat": {{fat2}}, "fiber": {{fiber2}}, "protein": {{protein2}}, "carbohydrates": {{carbohydrates2}}
 ...
]
 },
 {
 "name": "{{mealName2}}",
 "foods": [
 {"food_name": "{{foodName1}}", "amount": {{amount1}}, "energy_per_100g": {{energy1}}, "fat": {{fat1}}, "fiber": {{fiber1}}, "protein": {{protein1}}, "carbohydrates": {{carbohydrates1}}},
 {"food_name": "{{foodName2}}", "amount": {{amount2}}, "energy_per_100g": {{energy2}}, "fat": {{fat2}}, "fiber": {{fiber2}}, "protein": {{protein2}}, "carbohydrates": {{carbohydrates2}}
 ...
]
 },
 ...
]
 },
 ...
]
}
```

## B Prompt for Synthesising Nutrition Values of Foods and Ingredients for First Experiment

As a nutrition data analyst with extensive experience, your task is to provide JSON-formatted nutritional information for various ingredients, strictly adhering to USDA nutrition standards. This accurate and reliable data is critical for healthcare applications where precise nutrient information is essential for patient care.

Please format the nutritional data per 100 grams for each specified ingredient, focusing on energy, carbohydrates, protein, fats, and fiber. The JSON output must be clear and accurate to ensure seamless integration with healthcare applications.

**\*\*JSON Template:\*\***

```
```json
{
  "Ingredient1": {"Energy": "{energyValue1} kcal", "Carbs": "{carbsValue1} g", "Protein": "{proteinValue1} g", "Fats": "{fatValue1} g",
  "Fiber": "{fiberValue1} g"},
  // Continue this pattern until all ingredients are done.
}
```
```

ingredient list:  
[ingredient\_list]

## C GPT's Outputs, Structured as JSON Strings Including Nutrition Values

```
```json
{
  "Whole Wheat Pasta": {"Energy": "124 kcal", "Carbs": "25.77 g", "Protein": "4.48 g", "Fats": "0.87 g", "Fiber": "3.2 g"},
  "Bulgur Wheat": {"Energy": "83 kcal", "Carbs": "18.58 g", "Protein": "3.08 g", "Fats": "0.24 g", "Fiber": "4.5 g"},
  "Pasta": {"Energy": "131 kcal", "Carbs": "25.77 g", "Protein": "4.5 g", "Fats": "1.12 g", "Fiber": "1.3 g"},
  "Buckwheat": {"Energy": "343 kcal", "Carbs": "71.5 g", "Protein": "13.25 g", "Fats": "3.4 g", "Fiber": "10 g"},
  "Oats": {"Energy": "389 kcal", "Carbs": "66.27 g", "Protein": "16.89 g", "Fats": "6.9 g", "Fiber": "10.6 g"},
  "Spaghetti": {"Energy": "158 kcal", "Carbs": "31.57 g", "Protein": "5.8 g", "Fats": "1.04 g", "Fiber": "2.2 g"},
  "Gnocchi": {"Energy": "131 kcal", "Carbs": "25.77 g", "Protein": "4.5 g", "Fats": "1.12 g", "Fiber": "1.3 g"},
  "Focaccia": {"Energy": "249 kcal", "Carbs": "50.4 g", "Protein": "9.22 g", "Fats": "2.64 g", "Fiber": "2.1 g"},
  "White Potatoes": {"Energy": "77 kcal", "Carbs": "17.49 g", "Protein": "1.89 g", "Fats": "0.1 g", "Fiber": "2.2 g"},
  "Macaroni": {"Energy": "131 kcal", "Carbs": "25.77 g", "Protein": "4.5 g", "Fats": "1.12 g", "Fiber": "1.3 g"},
  "Pita": {"Energy": "275 kcal", "Carbs": "53.33 g", "Protein": "9.3 g", "Fats": "1.67 g", "Fiber": "2.7 g"},
  "Bread": {"Energy": "265 kcal", "Carbs": "49.69 g", "Protein": "8.82 g", "Fats": "3.15 g", "Fiber": "4.9 g"},
  "Rye": {"Energy": "338 kcal", "Carbs": "73.51 g", "Protein": "10.34 g", "Fats": "1.63 g", "Fiber": "15.1 g"},
  "Couscous": {"Energy": "112 kcal", "Carbs": "23.22 g", "Protein": "3.79 g", "Fats": "0.15 g", "Fiber": "1.3 g"},
  "Multigrain Bread": {"Energy": "250 kcal", "Carbs": "45.64 g", "Protein": "11.79 g", "Fats": "3.57 g", "Fiber": "7.1 g"},
  "Corn": {"Energy": "365 kcal", "Carbs": "74.26 g", "Protein": "9.42 g", "Fats": "4.74 g", "Fiber": "7.3 g"},
  "Tortillas": {"Energy": "218 kcal", "Carbs": "44.84 g", "Protein": "5.76 g", "Fats": "2.4 g", "Fiber": "2.8 g"},
  "Quinoa": {"Energy": "120 kcal", "Carbs": "21.3 g", "Protein": "4.4 g", "Fats": "1.92 g", "Fiber": "2.8 g"},
  "Barley": {"Energy": "354 kcal", "Carbs": "73.48 g", "Protein": "12.48 g", "Fats": "2.3 g", "Fiber": "17.3 g"},
  "Naan": {"Energy": "320 kcal", "Carbs": "53.33 g", "Protein": "9.33 g", "Fats": "5.33 g", "Fiber": "2.7 g"},
  "Sourdough": {"Energy": "230 kcal", "Carbs": "45.83 g", "Protein": "8.33 g", "Fats": "1.67 g", "Fiber": "3.3 g"},
  "Sweet Potatoes": {"Energy": "86 kcal", "Carbs": "20.12 g", "Protein": "1.57 g", "Fats": "0.05 g", "Fiber": "3 g"},
  "Pancakes": {"Energy": "227 kcal", "Carbs": "43.83 g", "Protein": "7.37 g", "Fats": "2.64 g", "Fiber": "1.6 g"},
  "Rice": {"Energy": "130 kcal", "Carbs": "28.73 g", "Protein": "2.69 g", "Fats": "0.28 g", "Fiber": "0.4 g"},
  "English Muffin": {"Energy": "211 kcal", "Carbs": "39.77 g", "Protein": "8.45 g", "Fats": "1.89 g", "Fiber": "2.7 g"},
  "Polenta": {"Energy": "83 kcal", "Carbs": "17.75 g", "Protein": "1.42 g", "Fats": "0.34 g", "Fiber": "1.8 g"},
  "Bagel": {"Energy": "250 kcal", "Carbs": "50.56 g", "Protein": "9.05 g", "Fats": "0.78 g", "Fiber": "1.8 g"},
  "Potatoes": {"Energy": "77 kcal", "Carbs": "17.49 g", "Protein": "1.89 g", "Fats": "0.1 g", "Fiber": "2.2 g"},
  "Brown Rice": {"Energy": "111 kcal", "Carbs": "23.51 g", "Protein": "2.14 g", "Fats": "0.9 g", "Fiber": "1.6 g"},
  "Millet": {"Energy":
    "378 kcal", "Carbs": "72.85 g", "Protein": "11.02 g", "Fats": "4.22 g", "Fiber": "8.5 g"}
}
```

D Example of a Meal Plan Generated by GPT-4-Turbo

Here's a 1-day personalized diet plan based on the provided profile and dietary preferences, aimed to help with bulking lean muscle while adhering to a balanced diet as guided by the Eatwell principles:

```
### **Day 1 Meal Plan**

#### **Breakfast**
- **Oatmeal with Mixed Berries and Chia Seeds**
  - Rolled oats: 80g
  - Fresh mixed berries (strawberries, blueberries): 100g
  - Chia seeds: 15g
  - Skimmed milk: 250ml

#### **Lunch**
- **Quinoa and Grilled Chicken Salad**
  - Quinoa: 100g
  - Grilled chicken breast: 150g
  - Mixed salad greens (spinach, arugula): 75g
  - Cherry tomatoes: 50g
  - Cucumber: 50g
  - Olive oil (dressing): 10ml
  - Lemon juice: 5ml

#### **Dinner**
- **Salmon with Sweet Potato and Steamed Broccoli**
  - Salmon fillet: 150g
  - Sweet potato: 200g
  - Broccoli: 100g
  - Olive oil: 5ml

#### **Snack 1**
- **Greek Yogurt with Honey and Almonds**
  - Greek yogurt: 150g
  - Honey: 10g
  - Almonds: 20g

#### **Snack 2**
- **Hummus with Carrot and Celery Sticks**
  - Hummus: 50g
  - Carrot sticks: 50g
  - Celery sticks: 50g

### **Nutritional Composition and Considerations**

- **Caloric distribution** is aligned with your goals: higher in protein for muscle building and sufficient carbohydrates for energy, with healthy fats.
- **Fiber intake** is achieved through fruits, vegetables, and whole grains.
- **Fluids**: Includes water intake from foods and recommends additional 1.5 to 2 liters of water throughout the day to meet hydration needs.

This plan integrates diverse food sources across all major food groups, emphasizing high-quality protein from both plant and animal sources, whole grains, a variety of fruits and vegetables, and healthy fats. It limits the intake of saturated fat, added sugars, and salt.
```

Here's the meal plan structured in the requested JSON format:

```
{
  "days": [
    {
      "day": "Day 1",
      "meals": [
        {
          "name": "Breakfast",
          "foods": [
            {
              "food_name": "Rolled oats",
              "amount": 80,
              "energy_per_100g": 389,
              "fat": 7,
              "fiber": 10,
              "protein": 17,
              "carbohydrates": 66
            },
            {
              "food_name": "Mixed berries",
              "amount": 100,
              "energy_per_100g": 57,
              "fat": 0.3,
              "fiber": 2,
              "protein": 0.7,
              "carbohydrates": 14
            },
            {
              "food_name": "Chia seeds",
              "amount": 15,
              "energy_per_100g": 486,
              "fat": 31,
              "fiber": 34,
              "protein": 17,
              "carbohydrates": 42
            },
            {
              "food_name": "Skimmed milk",
              "amount": 250,
              "energy_per_100g": 35,
              "fat": 0.1,
              "fiber": 0,
              "protein": 3.4,
              "carbohydrates": 5
            }
          ]
        },
        {
          "name": "Lunch",
          "foods": [
            {
              "food_name": "Quinoa",
              "amount": 100,
              "energy_per_100g": 120,
              "fat": 1.9,
              "fiber": 2.8,
              "protein": 4.4,
              "carbohydrates": 21.3
            },
            {
              "food_name": "Grilled chicken breast",
              "amount": 150,
              "energy_per_100g": 165,
              "fat": 3.6,
              "fiber": 0,
              "protein": 31,
              "carbohydrates": 0
            },
            {
              "food_name": "Mixed salad greens",
              "amount": 75,
              "energy_per_100g": 23,
              "fat": 0.4,
              "fiber": 1.2,
              "protein": 2,
              "carbohydrates": 3.3
            },
            {
              "food_name": "Cherry tomatoes",
              "amount": 50,
              "energy_per_100g": 18,
              "fat": 0.2,
              "fiber": 1.2,
              "protein": 0.9,
              "carbohydrates": 3.9
            },
            {
              "food_name": "Cucumber",
              "amount": 50,
              "energy_per_100g": 12,
              "fat": 0.1,
              "fiber": 0.5,
              "protein": 0.6,
              "carbohydrates": 1.9
            },
            {
              "food_name": "Olive oil",
              "amount": 10,
              "energy_per_100g": 884,
              "fat": 100,
              "fiber": 0,
              "protein": 0,
              "carbohydrates": 0
            },
            {
              "food_name": "Lemon juice",
              "amount": 5,
              "energy_per_100g": 29,
              "fat": 0.3,
              "fiber": 0.4,
              "protein": 0.4,
              "carbohydrates": 9
            }
          ]
        },
        {
          "name": "Dinner",
          "foods": [
            {
              "food_name": "Salmon fillet",
              "amount": 150,
              "energy_per_100g": 208,
              "fat": 13,
              "fiber": 0,
              "protein": 22,
              "carbohydrates": 0
            },
            {
              "food_name": "Sweet potato",
              "amount": 200,
              "energy_per_100g": 86,
              "fat": 0.1,
              "fiber": 3,
              "protein": 1.6,
              "carbohydrates": 20
            },
            {
              "food_name": "Broccoli",
              "amount": 100,
              "energy_per_100g": 34,
              "fat": 0.4,
              "fiber": 2.6,
              "protein": 2.8,
              "carbohydrates": 6.6
            },
            {
              "food_name": "Olive oil",
              "amount": 5,
              "energy_per_100g": 884,
              "fat": 100,
              "fiber": 0,
              "protein": 0,
              "carbohydrates": 0
            }
          ]
        },
        {
          "name": "Snack 1",
          "foods": [
            {
              "food_name": "Greek yogurt",
              "amount": 150,
              "energy_per_100g": 59,
              "fat": 0.4,
              "fiber": 0,
              "protein": 10,
              "carbohydrates": 3.6
            },
            {
              "food_name": "Honey",
              "amount": 10,
              "energy_per_100g": 304,
              "fat": 0,
              "fiber": 0,
              "protein": 0.3,
              "carbohydrates": 82
            },
            {
              "food_name": "Almonds",
              "amount": 20,
              "energy_per_100g": 576,
              "fat": 49,
              "fiber": 12.2,
              "protein": 21,
              "carbohydrates": 22
            }
          ]
        },
        {
          "name": "Snack 2",
          "foods": [
            {
              "food_name": "Hummus",
              "amount": 50,
              "energy_per_100g": 166,
              "fat": 9.6,
              "fiber": 4,
              "protein": 8,
              "carbohydrates": 14
            },
            {
              "food_name": "Carrot sticks",
              "amount": 50,
              "energy_per_100g": 41,
              "fat": 0.2,
              "fiber": 2.8,
              "protein": 0.9,
              "carbohydrates": 9.6
            },
            {
              "food_name": "Celery sticks",
              "amount": 50,
              "energy_per_100g": 16,
              "fat": 0.2,
              "fiber": 1.6,
              "protein": 0.7,
              "carbohydrates": 3
            }
          ]
        }
      ]
    }
  ]
}
```

This plan includes all necessary nutrients and considers hydration, variety, and balanced macronutrient distribution. Make sure to adjust portion sizes and ingredients based on actual energy needs and personal taste.

E Foods and Ingredient List

Sorting foods and ingredients into a Python list enables us to easily query the NutritionIX API to fetch nutritional information.

```
foods_and_ingredients_list = ['almond milk', 'brie', 'butter', 'camembert', 'cheddar',  
'cheese', 'coconut milk', 'cottage cheese', 'cream', 'feta', 'full cream milk', 'gouda', 'greek  
yogurt', 'ice cream', 'kefir', 'lactose free milk', 'mascarpone', 'milk', 'mozzarella', 'oat  
milk', 'paneer', 'parmesan', 'provolone', 'rice milk', 'ricotta', 'skim milk', 'sour cream',  
'soy milk', 'tofu', 'yogurt', 'apples', 'asparagus', 'avocado', 'bananas', 'berries', 'broccoli',  
'carrots', 'celery', 'cherries', 'cucumbers', 'eggplant', 'garlic', 'grapes', 'kale', 'lemons',  
'lettuce', 'limes', 'mangoes', 'onions', 'oranges', 'peaches', 'peppers', 'pineapple', 'plums',  
'pumpkin', 'spinach', 'sweetcorn', 'tomatoes', 'watermelon', 'zucchini', 'almond butter',  
'avocado oil', 'butter', 'canola oil', 'cashew butter', 'coconut oil', 'corn oil', 'cream cheese',  
'flaxseed oil', 'ghee', 'grapeseed oil', 'hazelnut oil', 'hemp oil', 'honey', 'jam', 'lard', 'mar-  
garine', 'marmalade', 'nutella', 'olive oil', 'peanut butter', 'peanut oil', 'pumpkin seed  
oil', 'sesame oil', 'soybean oil', 'sunflower oil', 'tahini', 'vegemite', 'vegetable oil', 'wal-  
nut oil', 'almonds', 'beans', 'cashews', 'chicken breast', 'chickpeas', 'cod', 'crab', 'duck',  
'eggs', 'ground beef', 'ham', 'lamb', 'lentils', 'lobster', 'mackerel', 'mussels', 'oysters', 'pea-  
nuts', 'pistachios', 'pork chops', 'salmon', 'sausages', 'scallops', 'shrimp', 'tempeh', 'tofu',  
'tuna', 'turkey', 'veal', 'walnuts', 'bagel', 'barley', 'bread', 'brown rice', 'buckwheat', 'bul-  
gur wheat', 'corn', 'couscous', 'english muffin', 'focaccia', 'gnocchi', 'macaroni', 'millet',  
'multigrain bread', 'naan', 'oats', 'pancakes', 'pasta', 'pita', 'polenta', 'potatoes', 'quinoa',  
'rice', 'rye', 'sourdough', 'spaghetti', 'sweet potatoes', 'tortillas', 'white potatoes', 'whole  
wheat pasta']
```